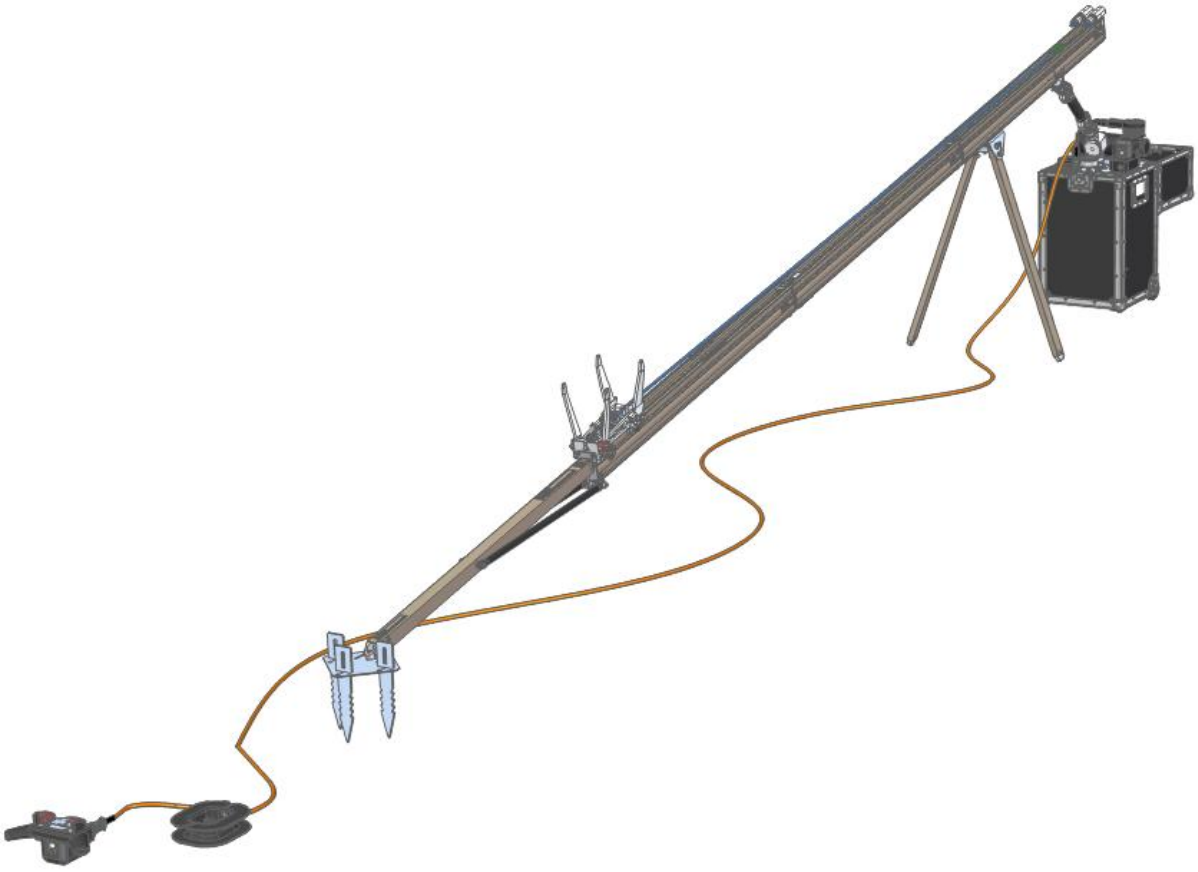
UAV pneumatic launcher PL-40
user manual



Tallinn 2026
ELI Military Simulations
REV 2.6

Contents

Introduction 3

1 Important safety instruction 3

2 Overview 4

3 Rail assembling procedure 7

4 Compressor unit 21

4.1 Overview 21

4.2 Operation 22

4.3 Setting the target pressure 22

4.4 Over discharge protection 23

5 Launching procedure 24

6 Disassembly procedure 27

7 Maintenance 28

7.1 Battery maintenance 28

7.2 Launcher maintenance 28

7.3 Pulling rope replacement 29

7.4 Wheel bearing replacement 34

8 Technical Specifications 37

9 Safety area schematics 40

10 Plane installation documentation in standard configuration 41

Introduction

The UAV pneumatic launcher is intended to accelerate UAV-s and other aircraft at takeoff with MTOW up to 30 kg and launch speed 25m/s. The launcher is light-weight, battery-operated and is quickly assembled. Device can be used for both puller and pusher propeller configuration aircraft.

The scope of this manual is to describe guidelines and procedures of the launcher assembly, disassembly and operation. These instructions must be strictly followed to ensure launch safety and trouble-free operation of the launcher.

1 Important safety instruction

Read these instructions carefully to avoid possible injury or property damage.



NEVER LAUNCH WITHOUT PAYLOAD as it will damage the device and surrounding persons!



NEVER LAUNCH WITHOUT SECURING LAUNCHERS REAR LEG TO THE GROUND
Backlash can move launcher and throw plane off it prematurely resulting in possible crash or other accident!



ALWAYS USE GLOVES AND GOGGLES when servicing the plane or the launcher!



ATTENTION! Before launching make sure that all team members are in safety area behind the launcher!



ATTENTION! Remove the carriage safety lock only just before the launch!



ATTENTION! Do not handle any pneumatic tubing until making sure that the valve of the main pressure tank is closed!



ATTENTION! Carriage has to be secured by the safety lock pin before opening the main valve!



WARNING! Open the main valve only before launch and close it after launch as soon as possible! Ignoring that may cause accidental sled or pressure release and lead to serious injuries or death!

2 Overview

All components are packed into two cases for easy transportation and handling (**Figure 2-2 & Figure 2-3**).

The launcher has following parts (Figure 2-1):

- Rear part of the rail (1) with rear support leg (2)
- Middle part of the rail (3)
- Front part of the rail (4) with front support legs (5) and pulling rope (6)
- Carriage (7)
- Ground anchors (8)
- Anchor disk (9) (The appearance may vary depending on the configuration)
- Hammer (10) (The appearance may vary depending on the configuration)
- Bag for ground anchors and hammer (11)
- Compressor unit (12)
- Trigger handle (13)
- Connection hose (14)
- 2 Makita-type 18V batteries (15)
- User manual (16)
- Transportation case for the rails (17)

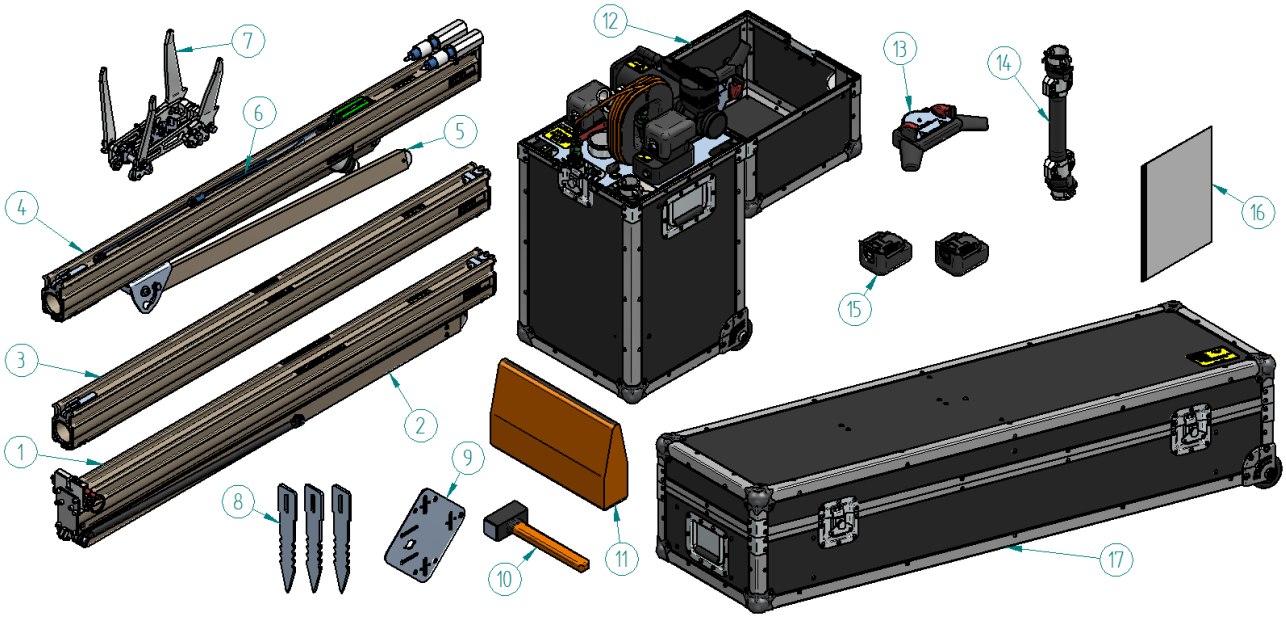


Figure 2-1 *Launcher kit parts (Design and equipment may vary depending on the configuration)*

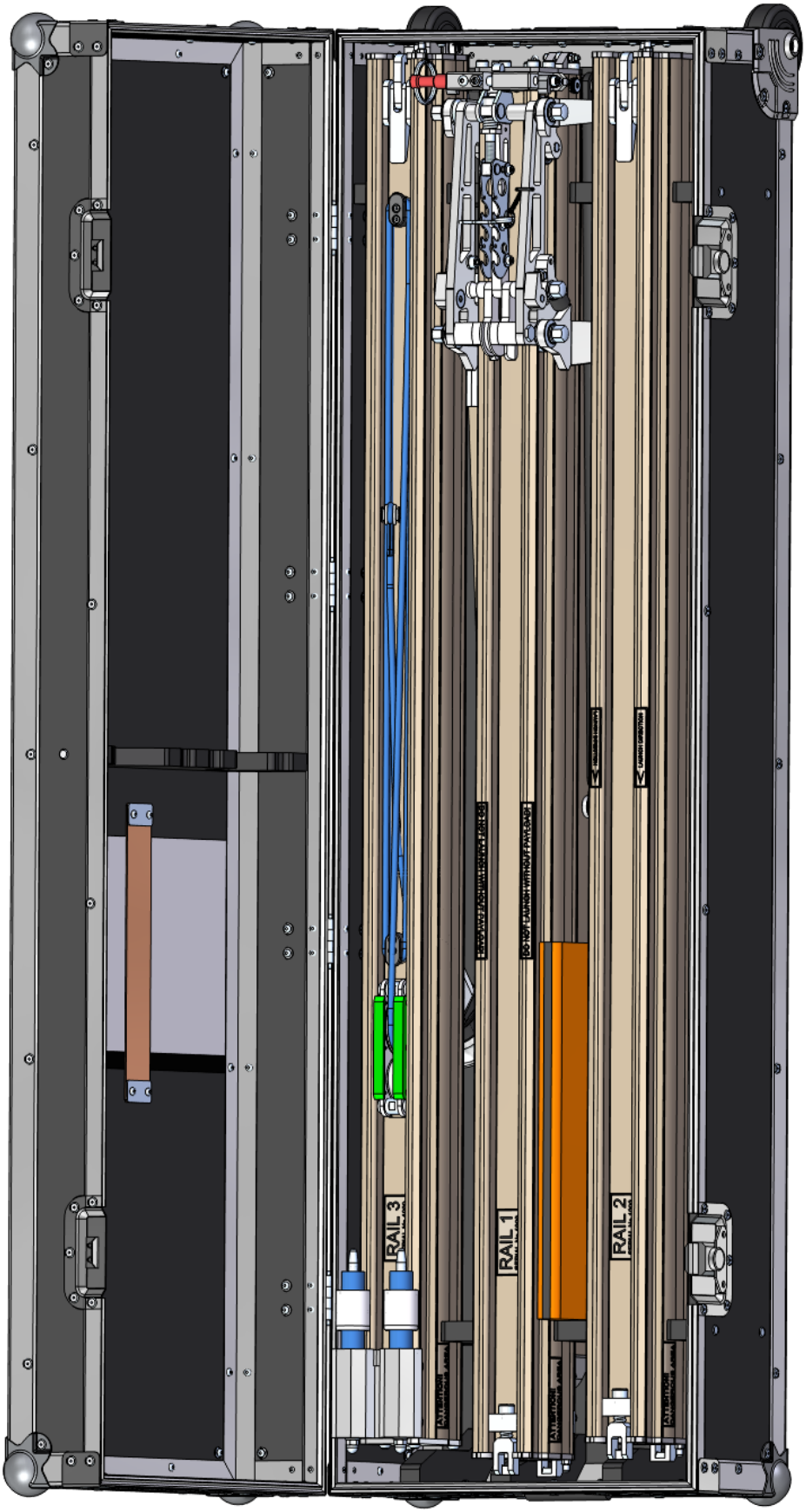


Figure 2-2 *Packed launcher rails*



Figure 2-3 Packed pneumatic system

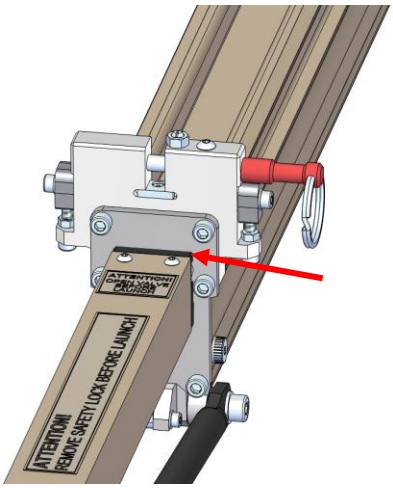
3 Rail assembling procedure

Before assembling the launcher choose the appropriate place and direction for takeoff
Determine wind direction and prefer to launch the pane directly into the wind.

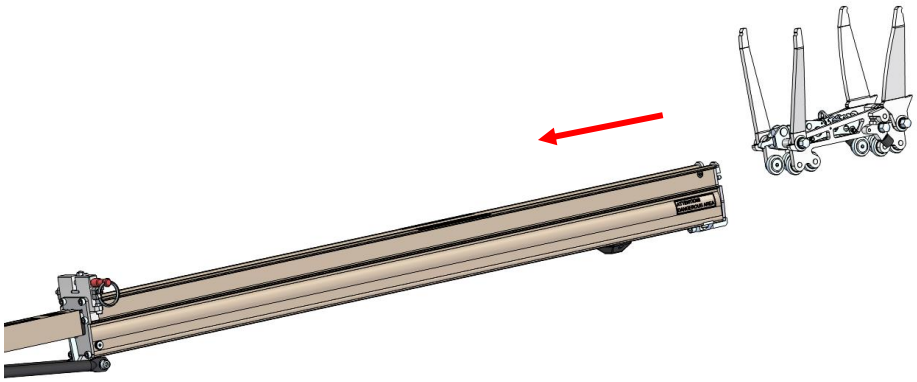
It is recommended to use an assistant when assembling the rails!



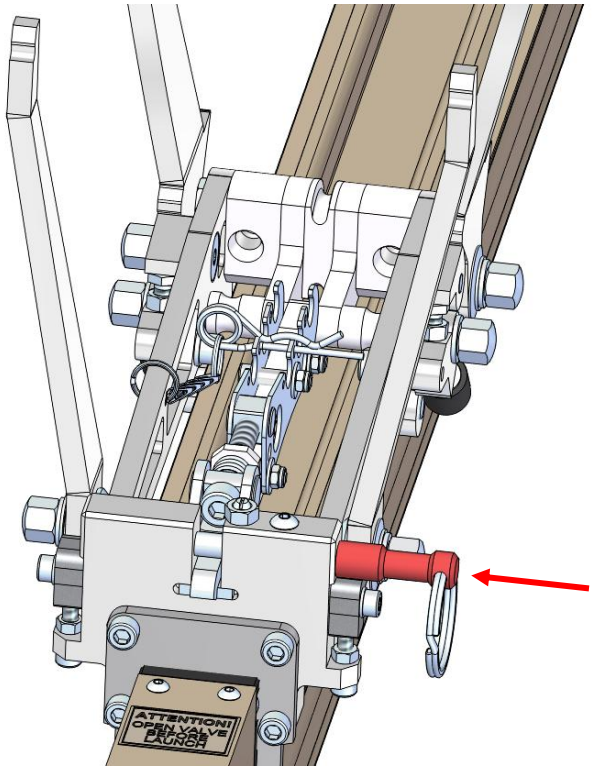
Step 1 *Take the rear rail and unfold the support leg.*



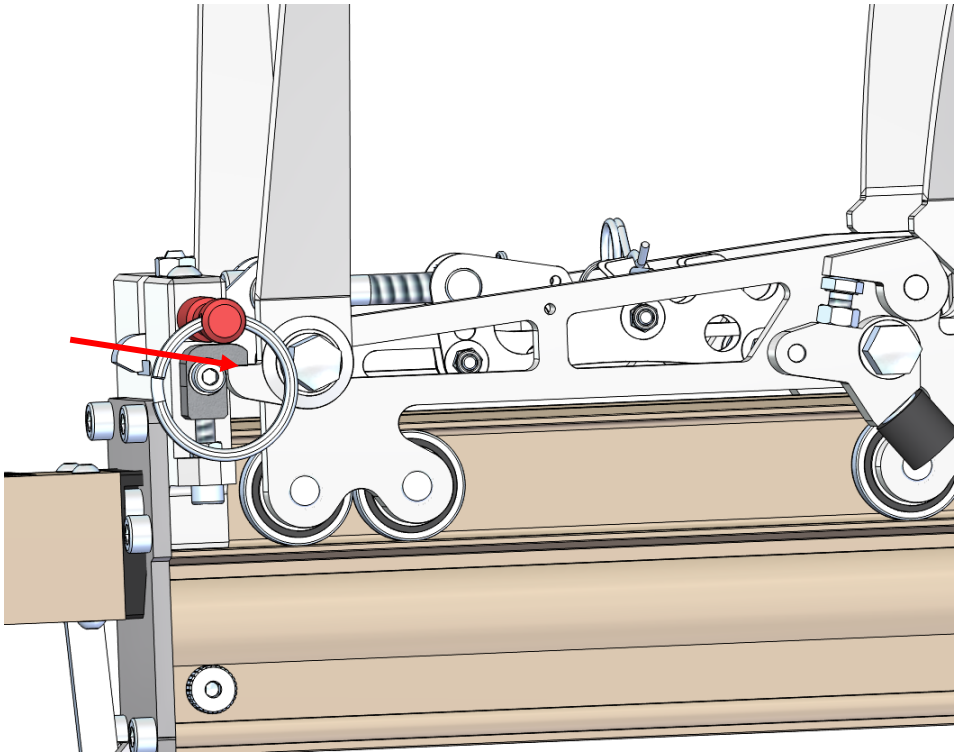
Step 2 *Make sure that the rear leg is properly seated at the end position.*



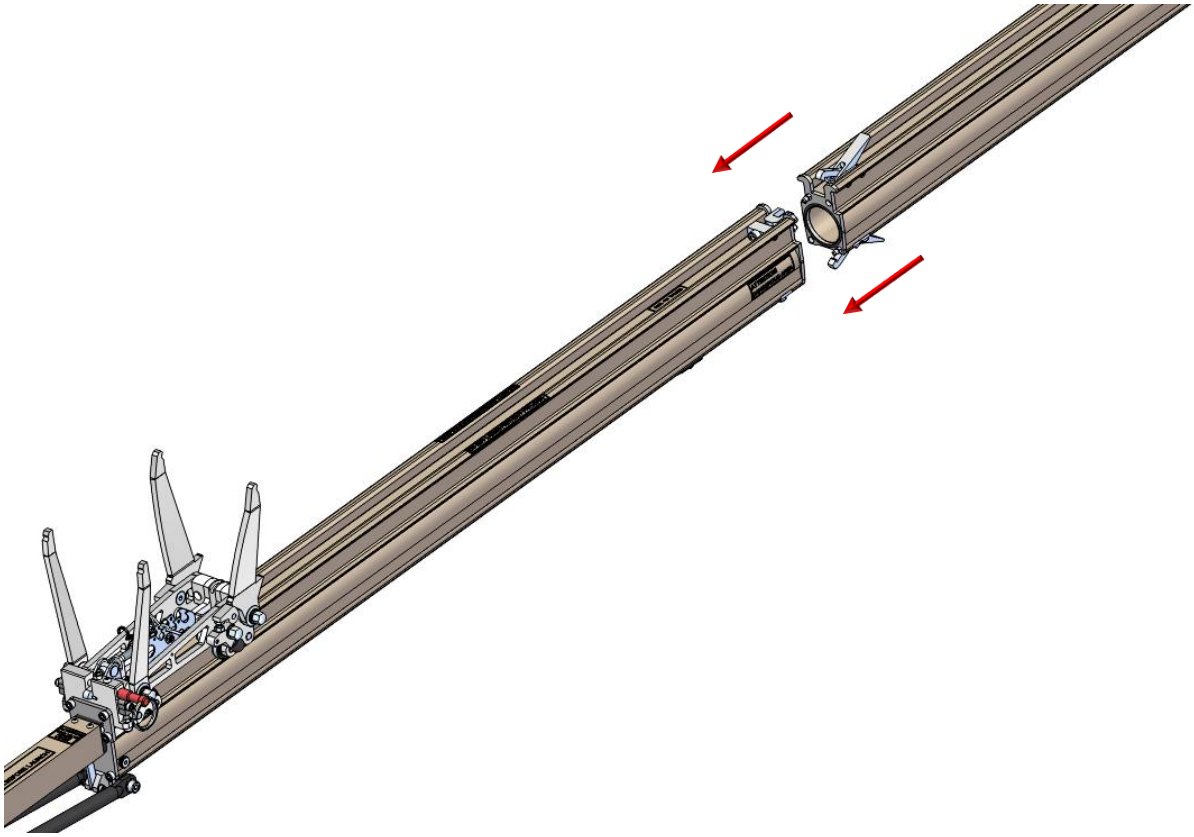
Step 3 *Place the carriage on rail and slide it to the end section.*



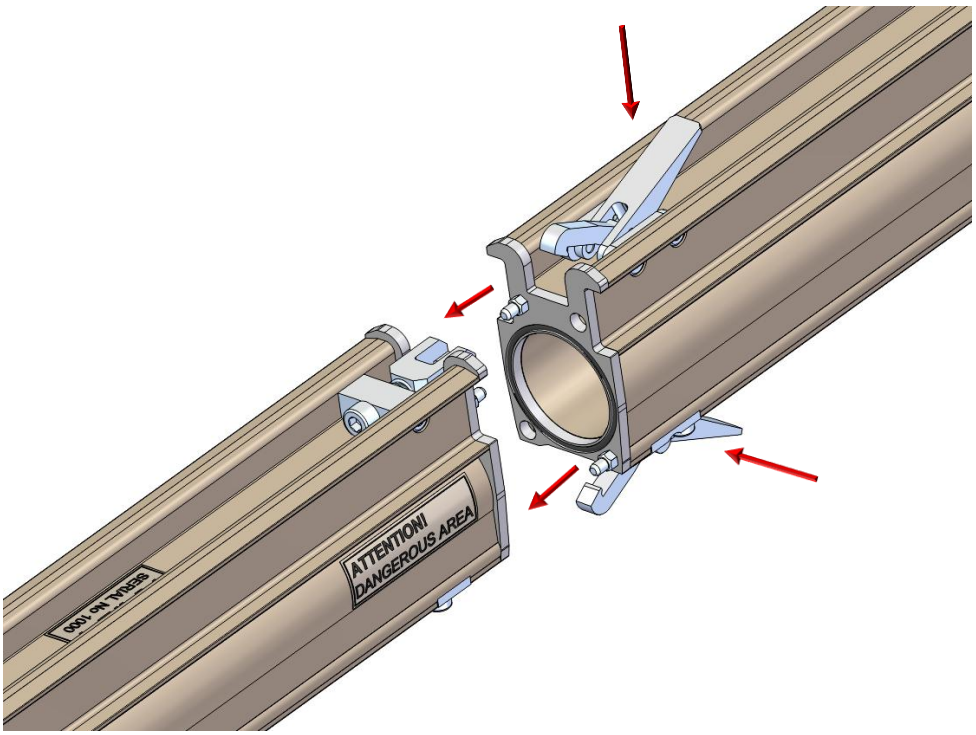
Step 4 *Secure the carriage with the safety lock pin.*



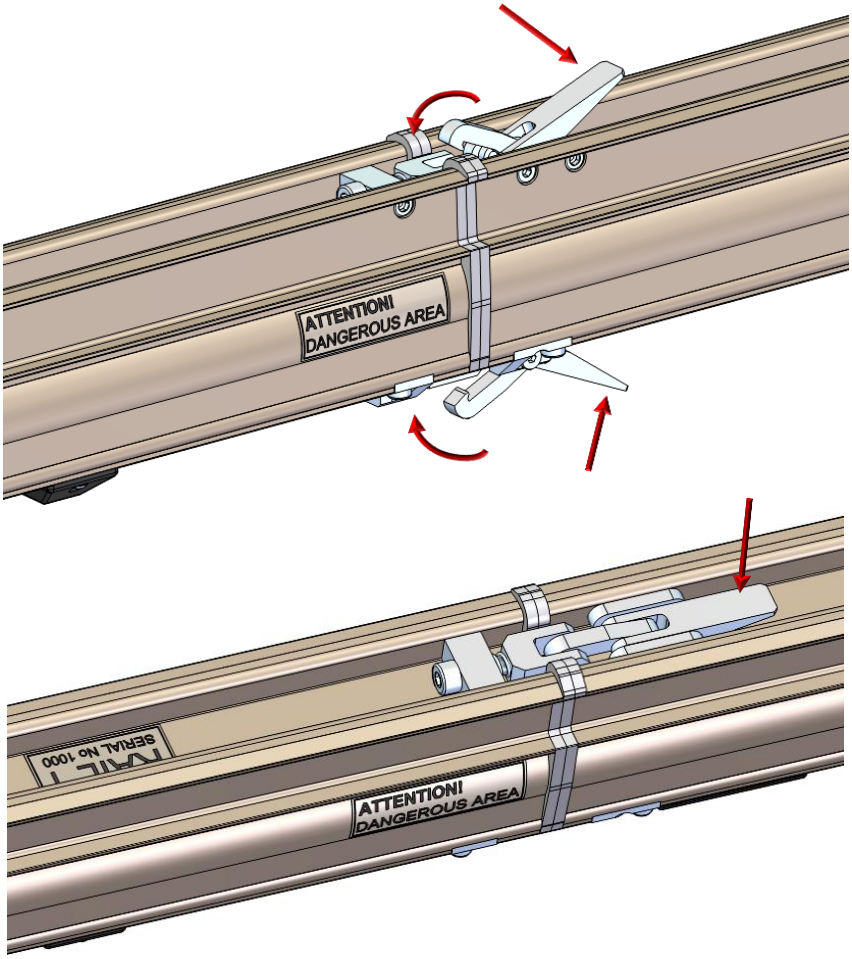
Step 5 *Make sure, that carriage support levers are in the upper position when locking the safety lock pin.*



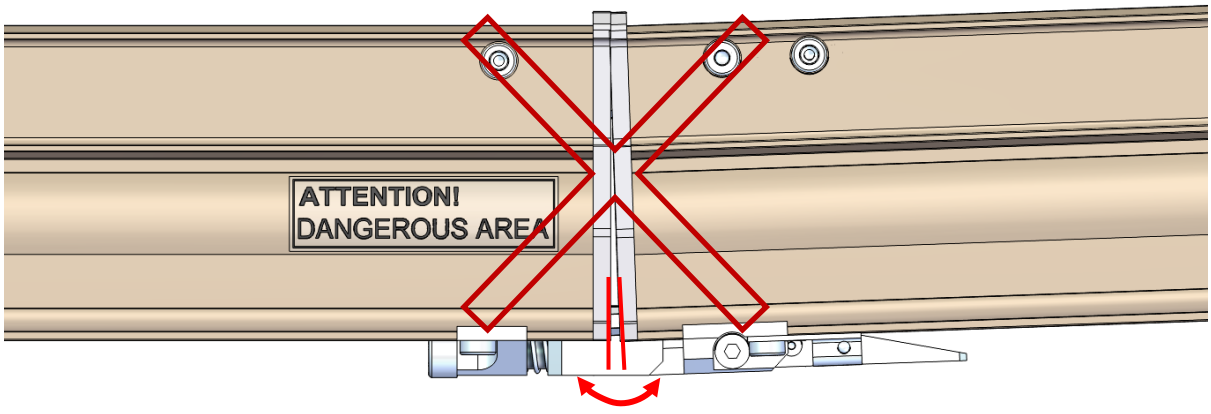
Step 6 *Line up the rear and the middle part of the rail.*



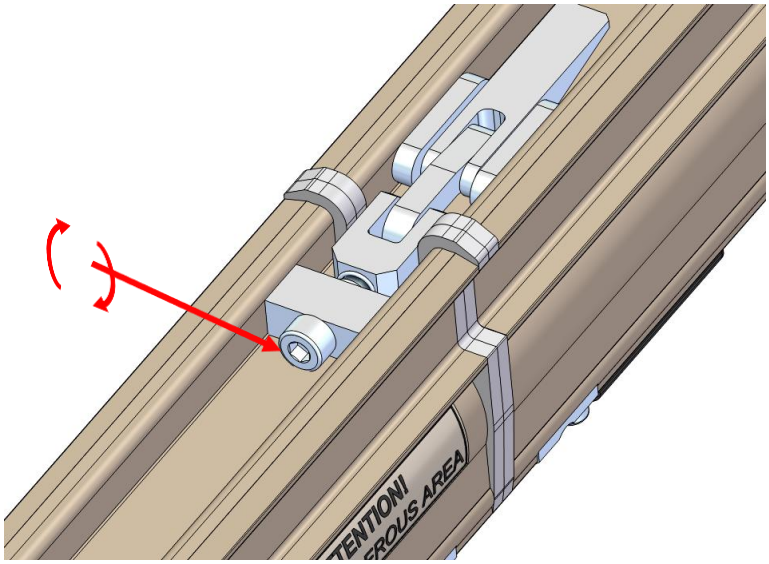
Step 7 *Release the clamps and merge the rails. Check that the O-rings are properly seated and that the end plates are clean.*

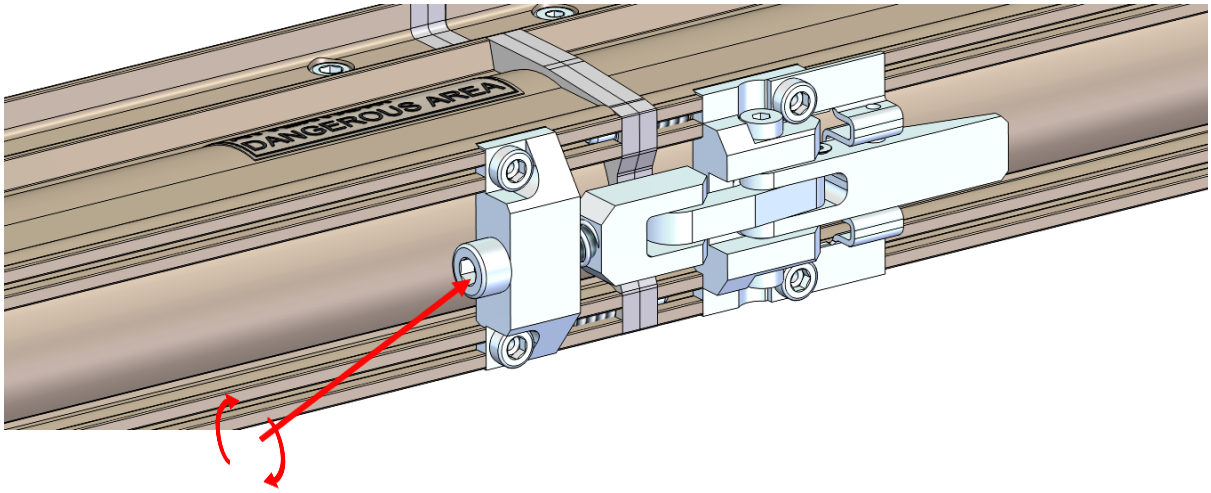


Step 8 *Lock the clamps and make sure that they are tightly secured in the locking position.*

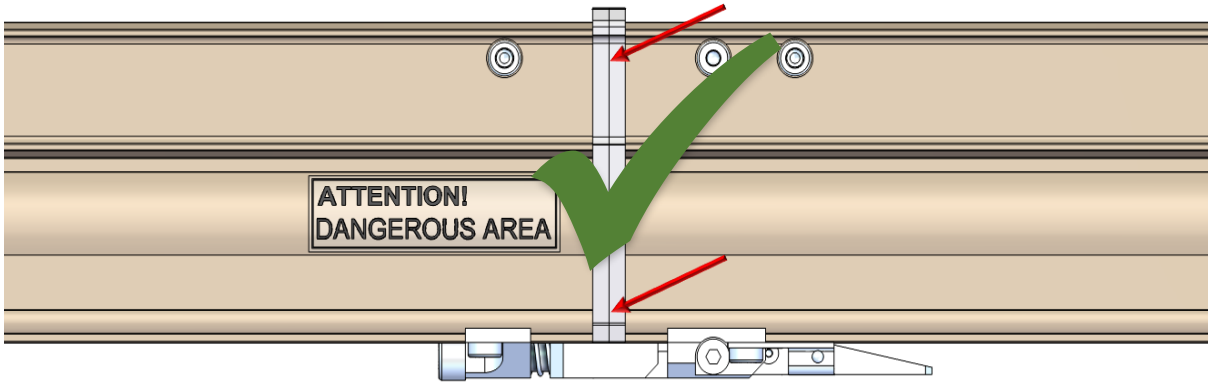


Step 9 *There should be no gaps or slacks between the tubes.*

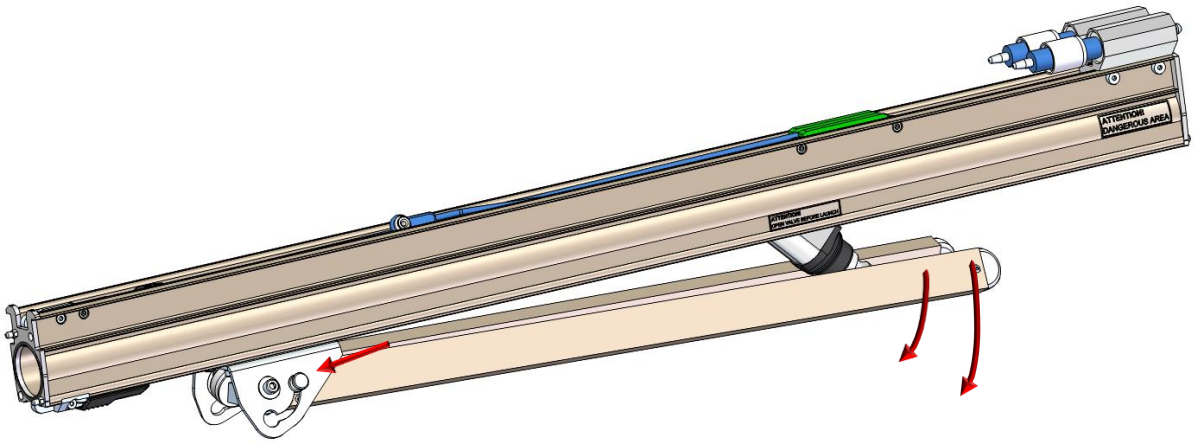




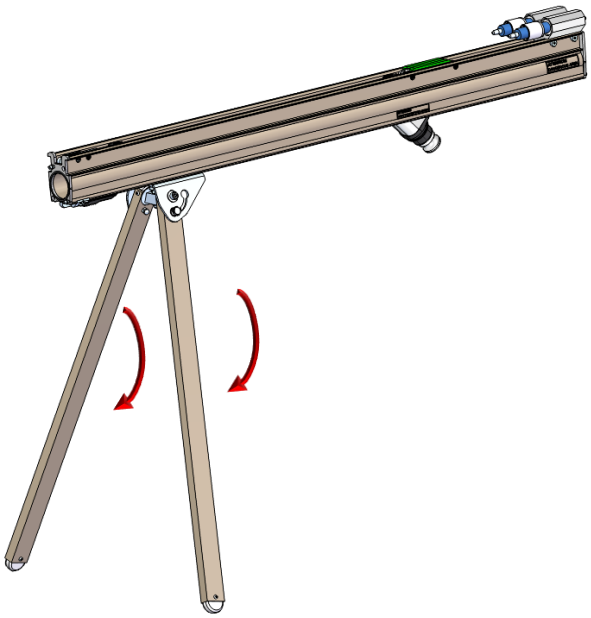
Step 10 *Use the clamp force adjustment screw to tighten the connection.*



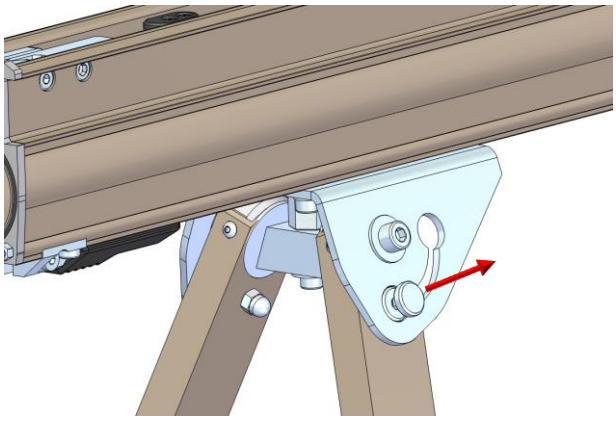
Step 11 *When adjusted correctly, the tubes should have no gaps and operator is able to open and close the clamps by hand force.*



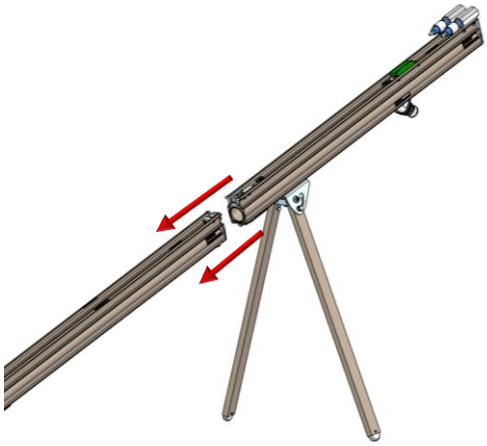
Step 12 *Push the locking pin to release the front support legs.*



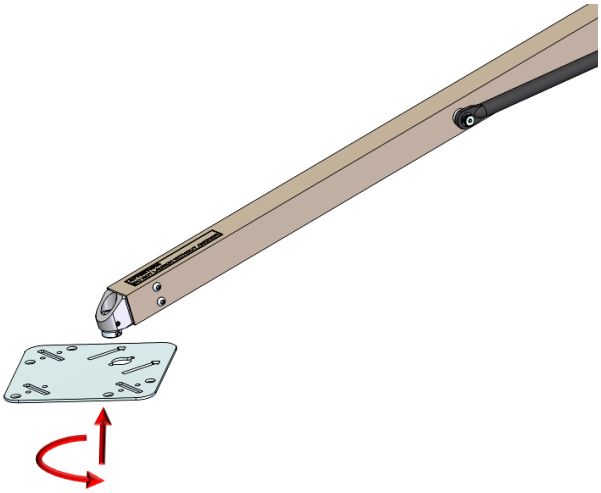
Step 13 *Move the legs into position.*



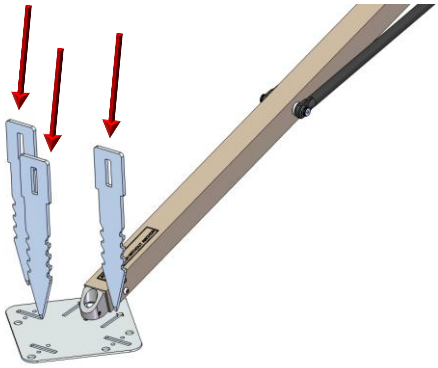
Step 14 *Make sure that the leg is secured and the lock has been engaged.*



Step 15 *Merge the front rail with the rest of the launcher assembly using the same method as in* ***Step 6*** *& Step 7.*



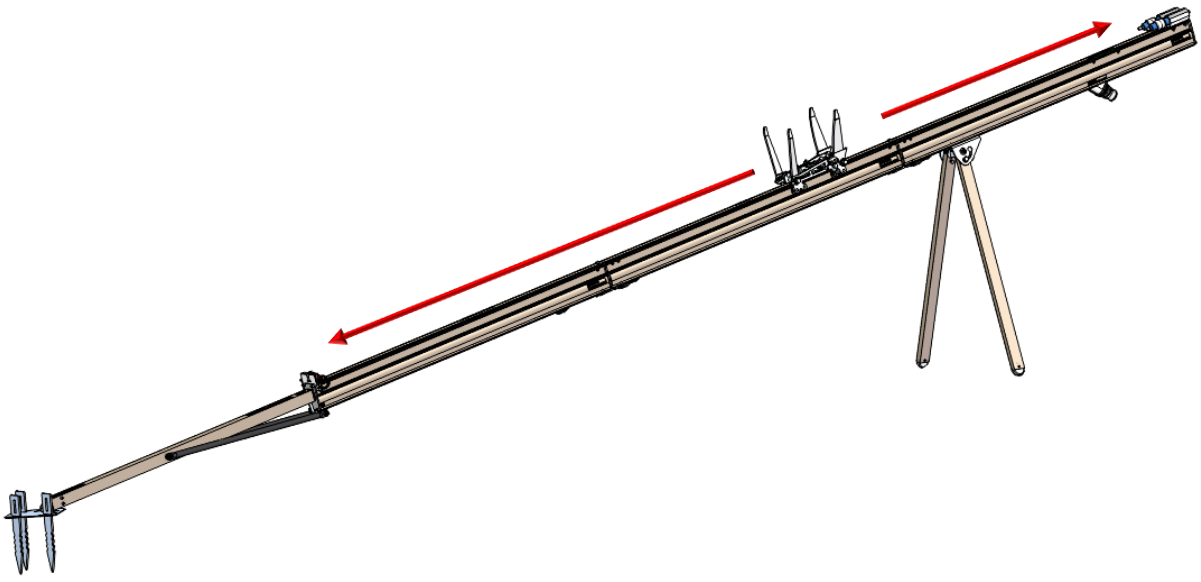
Step 16 *Mount the support disc turning it 90 degrees.*



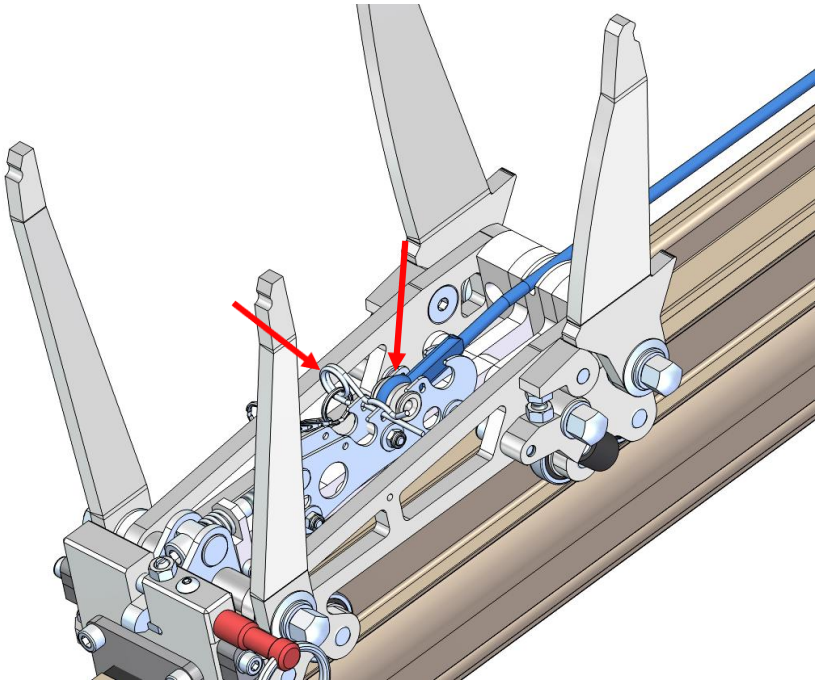
Step 17 *Make sure that the front rail support legs are in their end position before placing the anchors. The appearance of the anchor and support disc may vary depending on the configuration.*



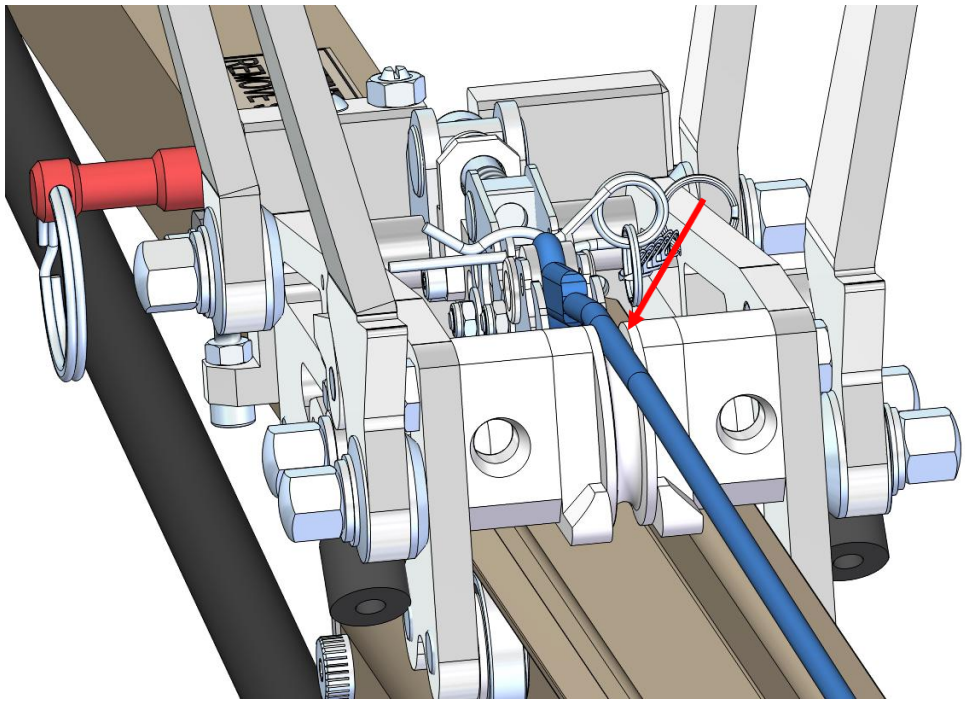
Step 18 *Secure the anchor using hammer or boot. Make sure that the soil is strong enough to withstand the launch forces. If not, the launch may fail and cause serious damage or injury. The catapult's anchoring to the ground must ensure that it cannot move during launch. The catapult operator is responsible for making sure the catapult is properly secured to the ground, using additional anchors in soft soil, etc.*



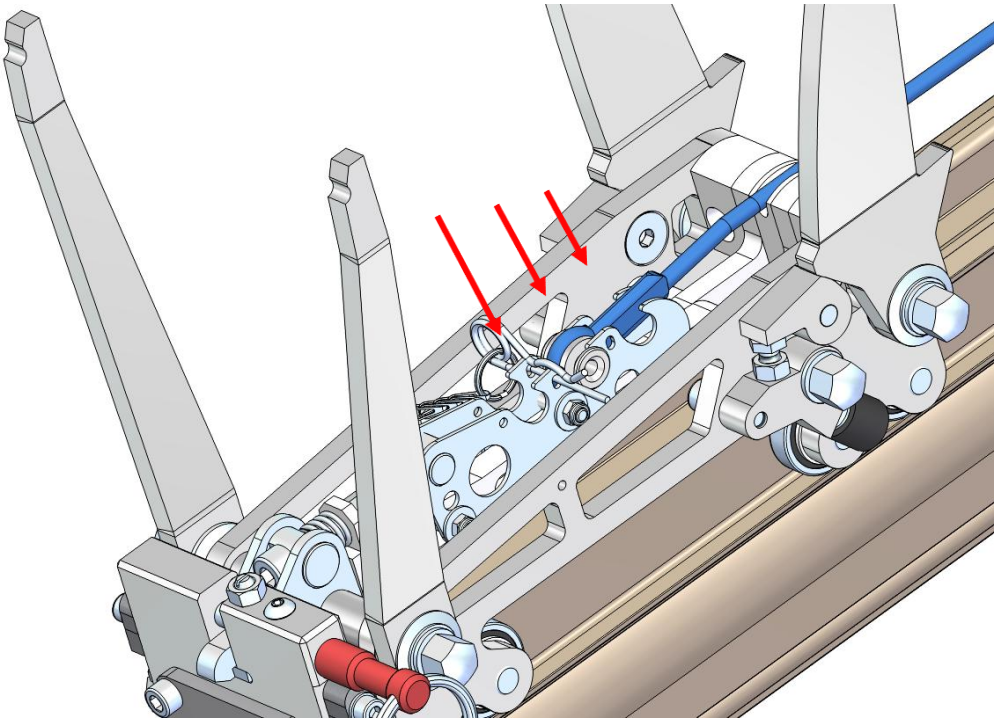
Step 19 *By moving the carriage back and forth make sure that the carriage is moving smoothly and have no jams or obstacles. After completion move the carriage back to the rear end of the rail and secure it with safety lock pin.*



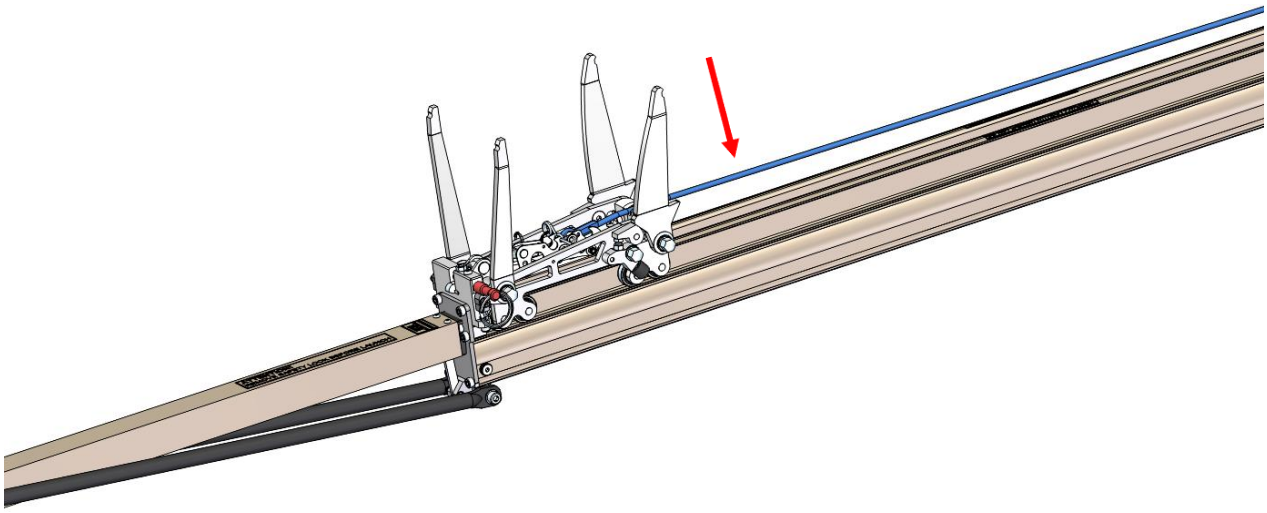
Step 20 *Connect the pulling rope to the furthest socket that the rope reaches and secure it with the safety pin. Make sure, that the rope is undamaged and clean, if not, then the rope must be replaced* ***(see paragraph 7.3 on page 29).***



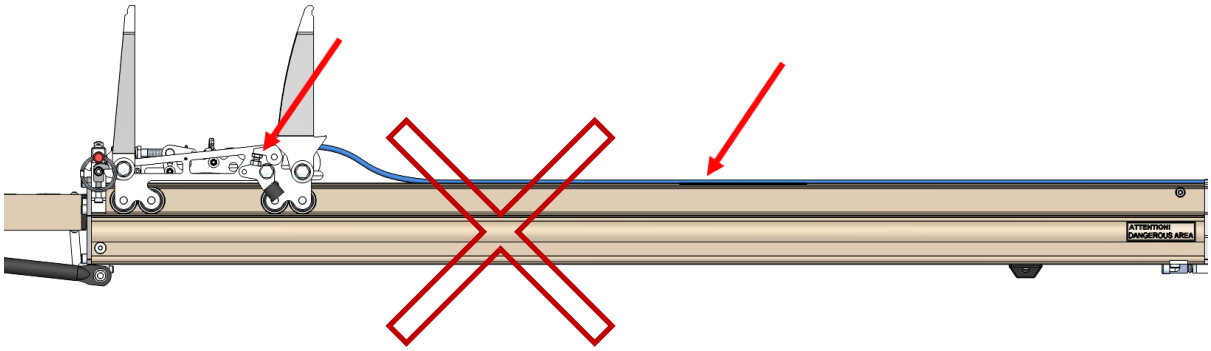
Step 21 *Make sure that the rope is passing the groove as above.*



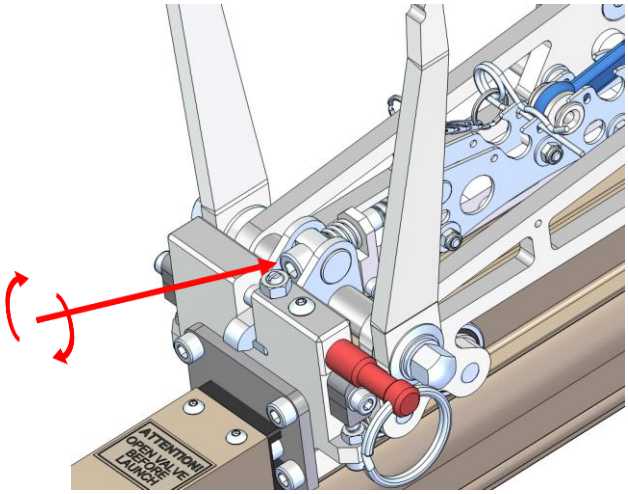
Step 22 *There are three slots for attaching and are used to compensate the length changes of the pulling rope.*



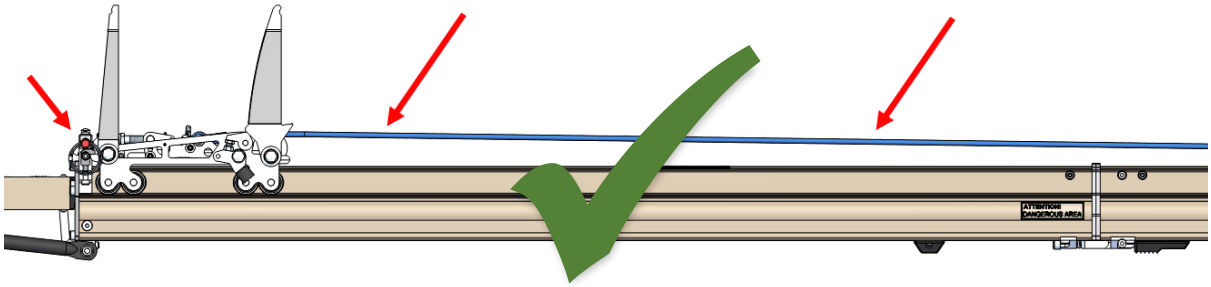
Step 23 *Attached rope must have enough tension. Make sure to check it before every launch. If the tension cannot be achieved then the rope must be replaced as shown on* ***(paragraph 7.3).***



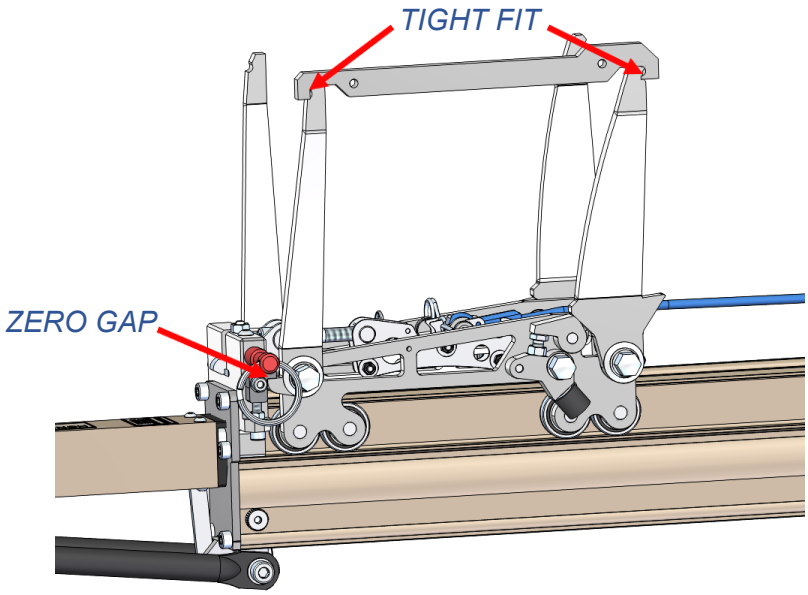
Step 24 *It is not recommended to launch with an unregulated pulling rope. If it is not possible to achieve the correct rope tension, it is important to ensure that the rope moves freely along the rail and that nothing obstructs its movement. A loose pulling rope may create a strong shock load at the moment of launch, which could result in excessive acceleration of the airframe.*



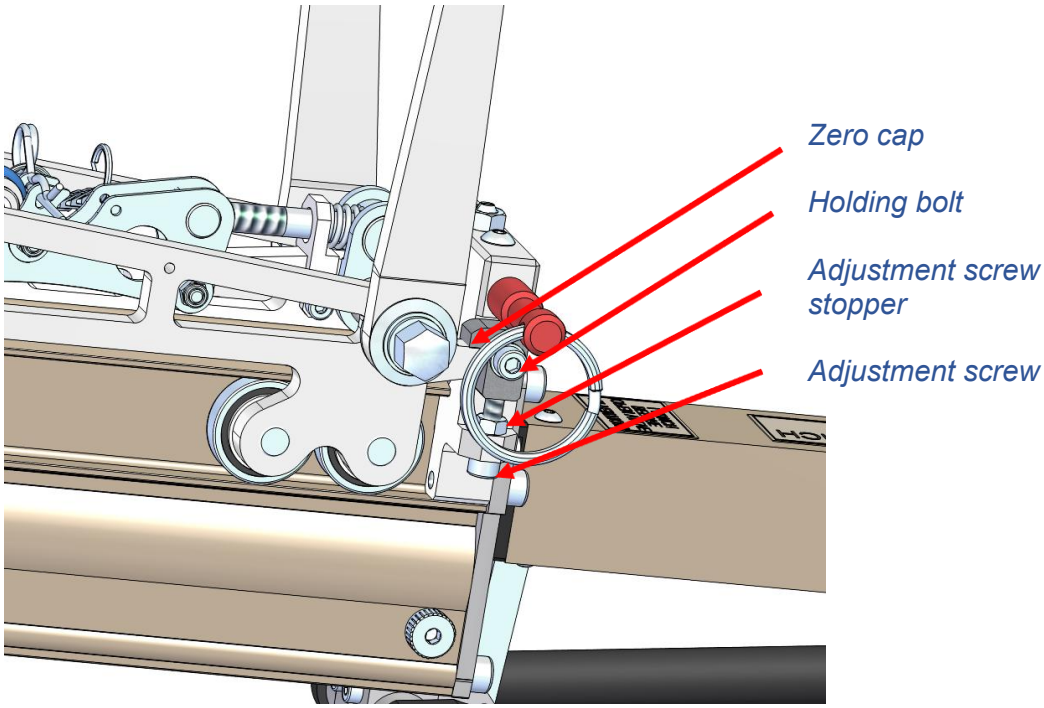
Step 25 *Use the available three rope slots and\or the regulating screw to obtain the proper pulling rope tension.*



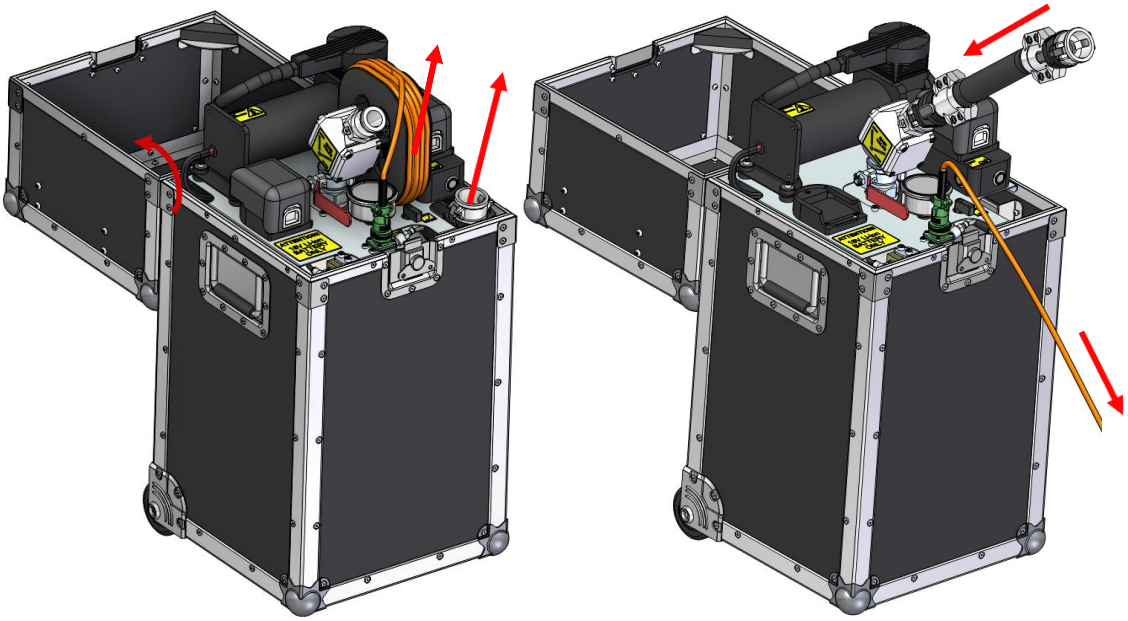
Step 26 *Properly adjusted rope should not be leaning on the rail but in the same time the tension should not be able to disengage the locking mechanism in the rear of the carriage. The carriage locking mechanism should engage only when the safety pin is removed and rope is pulled by a pressurized piston.*



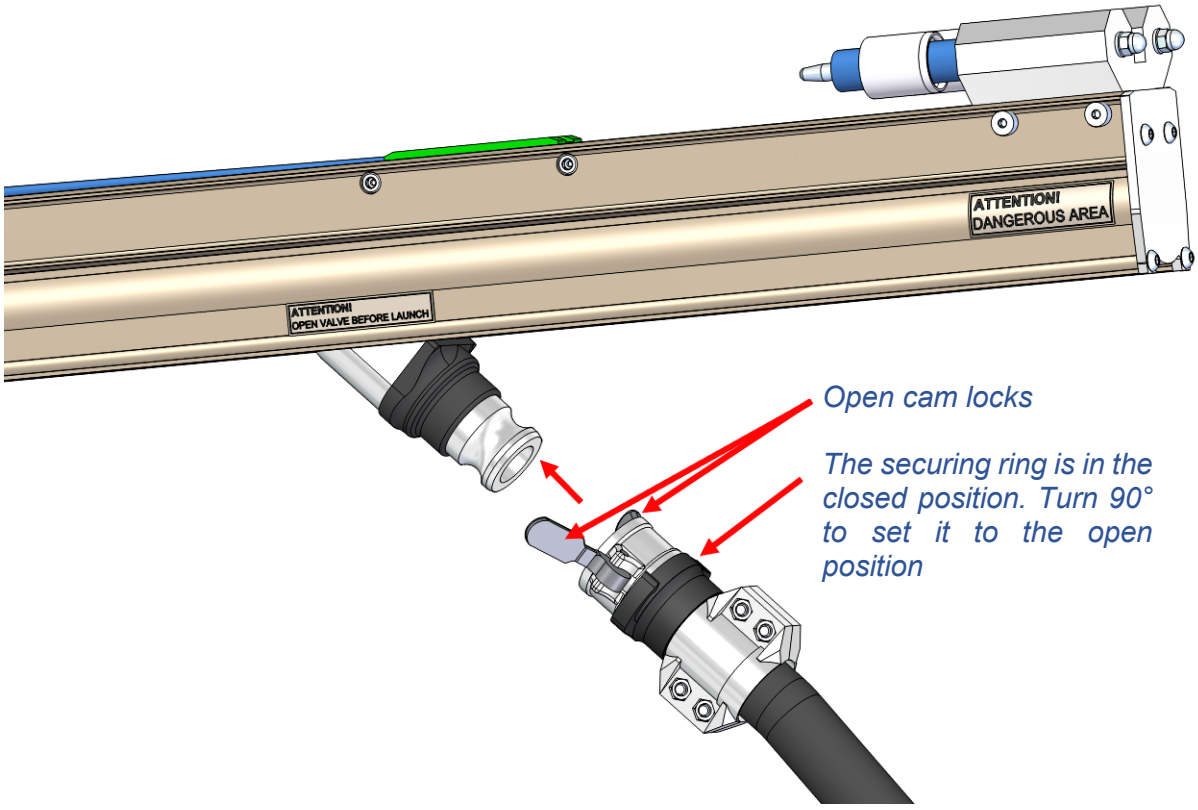
Step 27 *Make sure that levers are tightly inserted into plane's connection hooks. There should be no gap between levers lock pads. No lever movement should be present when the plane engine is throttled up and down. Inside the launcher are pneumatic safety mechanisms for shutting down the inlet pressure when piston over-acceleration occurs. Loose levers will momentarily over-accelerate the piston engaging the safety valve and the launch will fail.*



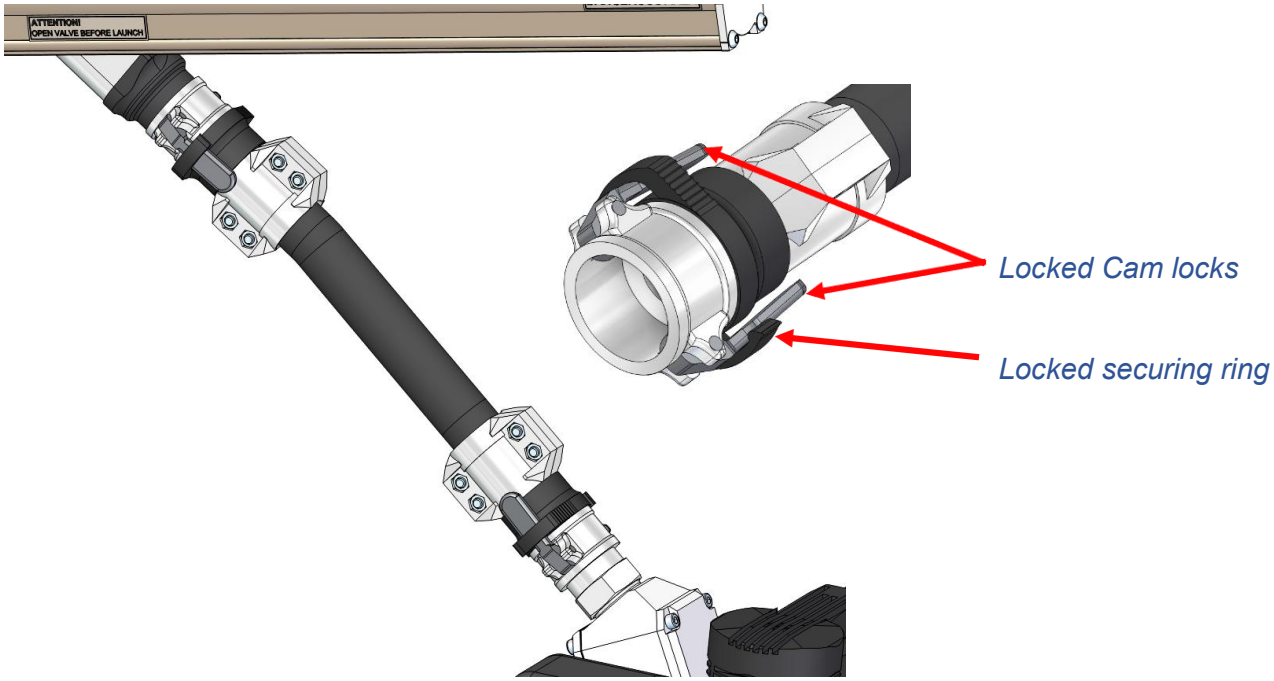
Step 28 *For adjusting the gap release the holding bolt and adjusting screw stopper. Change the height of the lock pad with the adjusting screw until there is no gap between lock bad and lever. Tighten the bolt and adjusting screw stopper. Same procedure must be completed for both levers.*



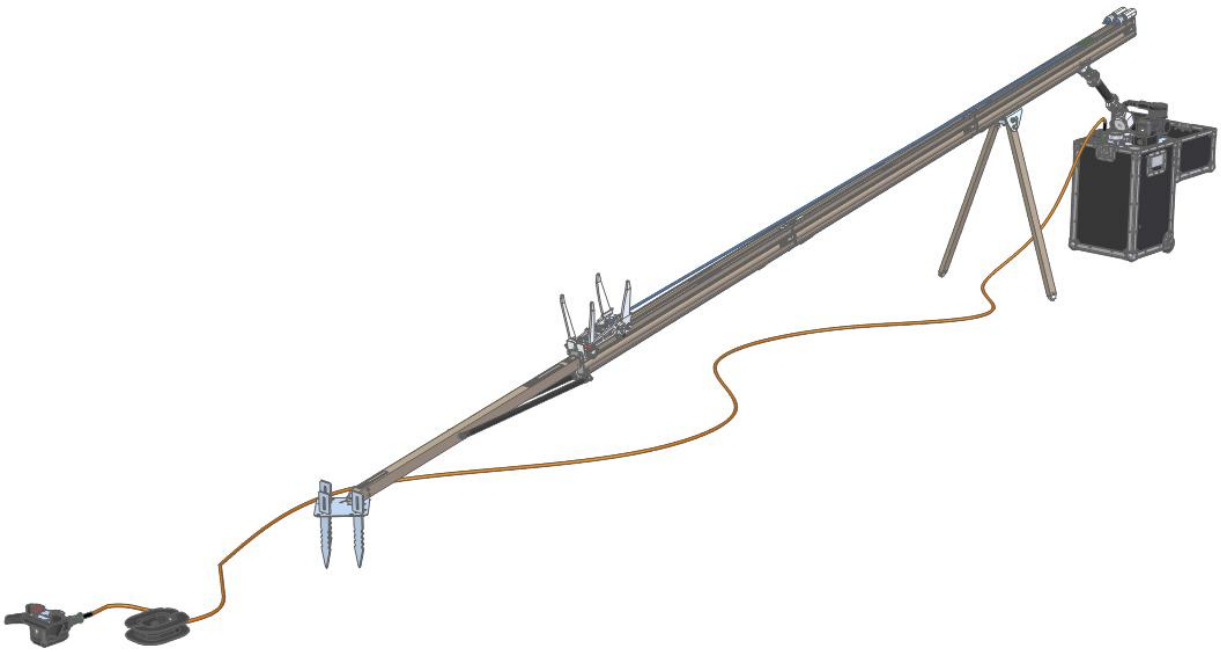
Step 29 *Open the lid of the compressor unit. Unwind the trigger handle and attach the connection hose.*



Step 30 *Connect one end of the connection tubing to the launcher inlet port and the other end of the tubing to the tank-side pressure connector. Before connecting the hoses, loosen the cam lock securing ring and the cam locks.*



Step 31 *After connecting both ends of the connection hoses, ensure that the cam locks are locked and that the cam lock secure ring is preventing the cam locks from opening*



Step 32 *Move the remote trigger unit to the determined position behind the rails. The launcher is now assembled and ready for start.*

4 Compressor unit

4.1 Overview

The catapult has battery operated air compressor with compressed air reservoir, electronic controller, and pneumatically operated main valve with remote trigger, pressure release valve, pressure gauge and safety valve.

The air compressor is operated by electronic controller that enables setting desired target pressure and protects system from overpressure and battery over discharge. The controller has ON/OFF switch with green LED illumination and SET button with red LED illumination.

The main parts of the compressor unit are shown in **Figure 4-1**:

1 – Compressor
2 – Tank side pressure connector
3 – Remote controlled release valve
4 – Main reservoir outlet valve
5 – Spare battery slot
6 – Main tank
7 – Main battery
8 – Pressure gauge
9 – Main battery slot
10 – Power ON/OFF switch
11 – Power switching circuit
12 – SET button
13 –Pressure connecting hose
14 – Tank draining valve
15 – Trigger handle
16 – Trigger handle cord spool
17 – Spare air filters for the compressor
18 – External filling port DIN 8 mm PCP

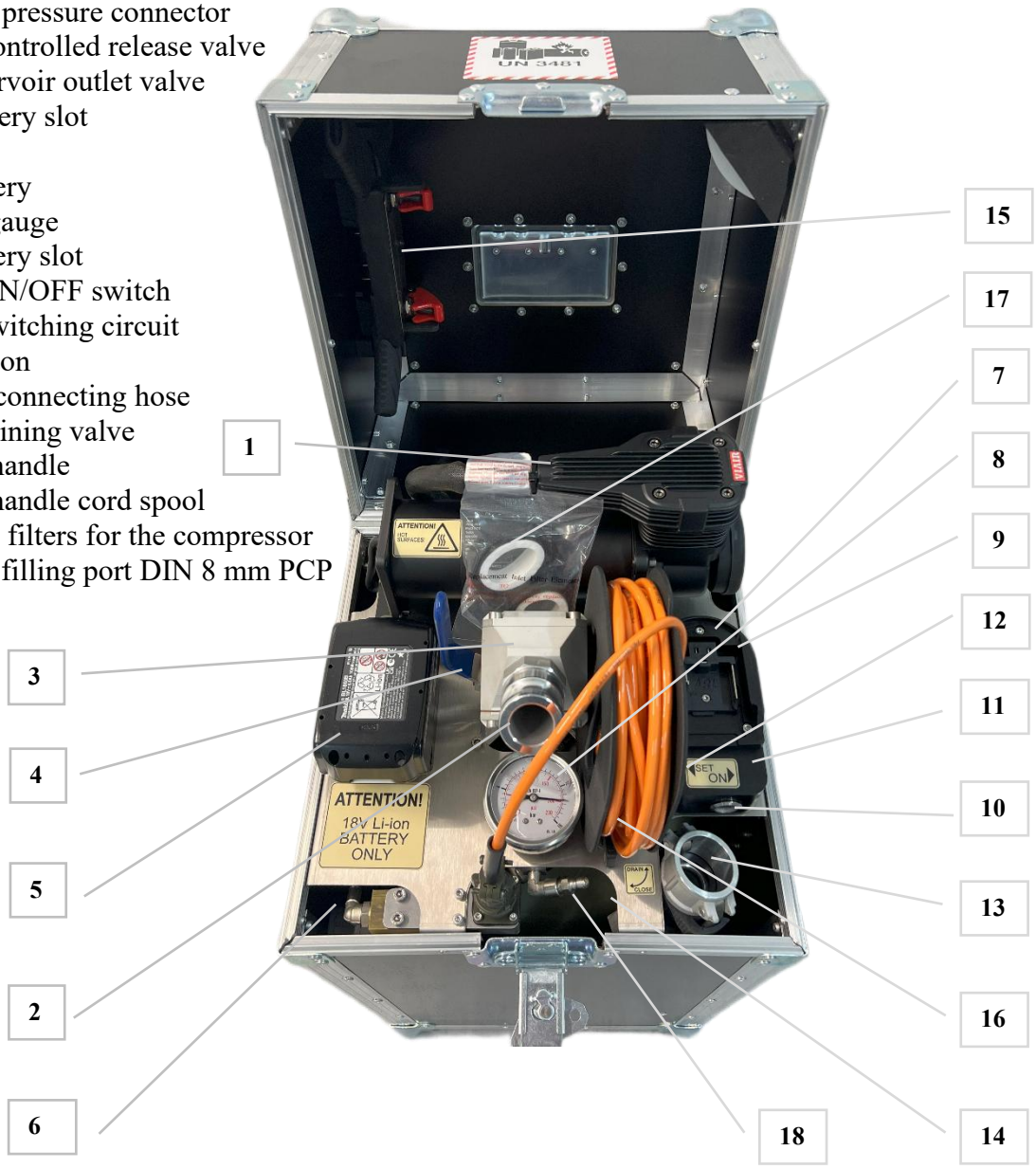


Figure 4-1 *Main parts of the compressor unit*

4.2 Operation

The controller is turned on by pushing ON/OFF switch to ON position. The controller checks battery voltage and pressure inside air reservoir. If the battery voltage is higher than cutoff voltage, ON LED is illuminated and if air pressure is lower than preset target pressure, the controller turns on the compressor and lights SET LED up. When target pressure is reached the controller stops the compressor. The controller hold determined pressure with 0.05 bar hysteresis and the catapult is ready for launch. After launch the controller charges the reservoir to the target pressure again.

If during initial check up the pressure in the reservoir is higher than target pressure, the compressor stays switched off and the SET LED flashes.

Table 1 Compressor unit LED states (illuminated push-buttons)

PWR LED	SET LED	MOTOR STATE	SYSTEM STATUS
○	○	**OFF**	Turned OFF
●	●	**OFF**	Target pressure reached
●	✹	**OFF**	Pressure in the tank is over the target pressure
●	●	**ON**	Pressurizing to target pressure
●	○	**ON**	SET MODE (Pressurizing tank)
●	○	**OFF**	SET MODE (Maximum pressure reached to 14 bar)
✹	**X**	**OFF**	Battery voltage too low

● - LED on ○ - LED off ✹ - LED blinking **X** - Any state

ATTENTION! Before launching the plane always check the pressure on the manometer, making sure, that the target pressure is achieved.

4.3 Setting the target pressure

To set or change target pressure the controller must be started in set mode. To enter SET MODE turn the device off, hold SET button pressed and turn it on again and release SET button. The controller enters SET MODE, turns SET LED off and starts the compressor. Monitor the pressure gauge and press the SET button when desired pressure is reached.
The controller stops the compressor, writes the new target pressure to the non-volatile memory, lights the SET LED up and resumes the normal operation.

NB! Make sure that the pressure in the reservoir is lower than the new target pressure before entering SET MODE. Use pressure release valve if necessary.

If the controller is in the SET MODE and the maximum pressure 14 bar is reached before SET button is pressed the compressor stops. If now the SET button is pressed then this maximum pressure is saved as new target pressure.

By turning off the controller before SET button is pressed, the set mode is canceled and previous target pressure remains in the memory.

4.4 Over discharge protection

The controller protects battery from destructive over discharge. When the battery voltage is fallen to or is below 14,5V, controller turns the compressor off and keeps ON LED flashing. When the battery recovers and voltage rises over 16V, the controller turns, after 5 a second delay, the compressor on again. It enables to get all remaining charge out of the battery. However, it is recommended to change the battery immediately after the protection engages.



ATTENTION! Do not handle any pneumatic tubing until making sure that the valve of the main pressure tank is closed!



ATTENTION! Carriage has to be secured by the safety lock pin before opening the main valve!



WARNING! Open the main valve only before launch and close it after launch as soon as possible! Ignoring that may cause accidental sled or pressure release and lead to serious injuries or death!

5 Launching procedure



WARNING!
Never use launcher without the plane! It can damage the device!



ATTENTION! It is user responsibility to maintain relationship between plane mass and determined launch pressure so, that maximum launch velocity 25 m/s is not exceeded. Consult Table 2 in Technical Specifications section.

For launching the plane perform the following steps:

1. Unpack and assemble the rail **(Chapter 3, on page 7)**
2. Make sure, that the carriage is in its initial position and secured with safety lock pin.
3. Make sure that the pulling rope is attached to the carriage and tensioned properly.

4. Make sure, that the compressor unit ON/OFF switch **(10)** is in OFF position and the main reservoir outlet valve **(4)** is in closed position.

5. Install pressure hose connecting main valve outlet and the launcher inlet **(13)**. Make sure that the clamps are perfectly locked and securing ring also locked **Step 31.**

6. Insert a fully charged battery cartridge into its connection slot **(9)** on the compressor unit.

7. Turn the automatic ON/OFF switch **(10)** ON. The compressor starts working and stops automatically when the pressure in the reservoir reaches preset value (up to 14bar). It can take up to 8 minutes until the maximum working pressure is achieved. The value of the determined pressure can be regulated via SET MODE **(Chapter 4.3, on page 22)**.

8. When the determined pressure is achieved, compressor stops. Check the pressure manometer on the pressure regulation valve to assure that the determined pressure is achieved. It is normal for the compressor to switch on from time to time for short periods of time to keep requested pressure.

9. If the compressor stops before the determined pressure is achieved the battery may be empty. Check the indication LED and change battery if necessary.

10. Place the plane on the launcher so that the two forward support levers of the carriage are inserted into their slots first, then lower the tail of the plane so that the rear pair of support levers are inserted into their slots **(Figure 5-1)**. Make sure that all four support levers are correctly in their positions and the plane sits evenly. Ensure that all drone-side hardpoints are straight, parallel, and securely tightened.

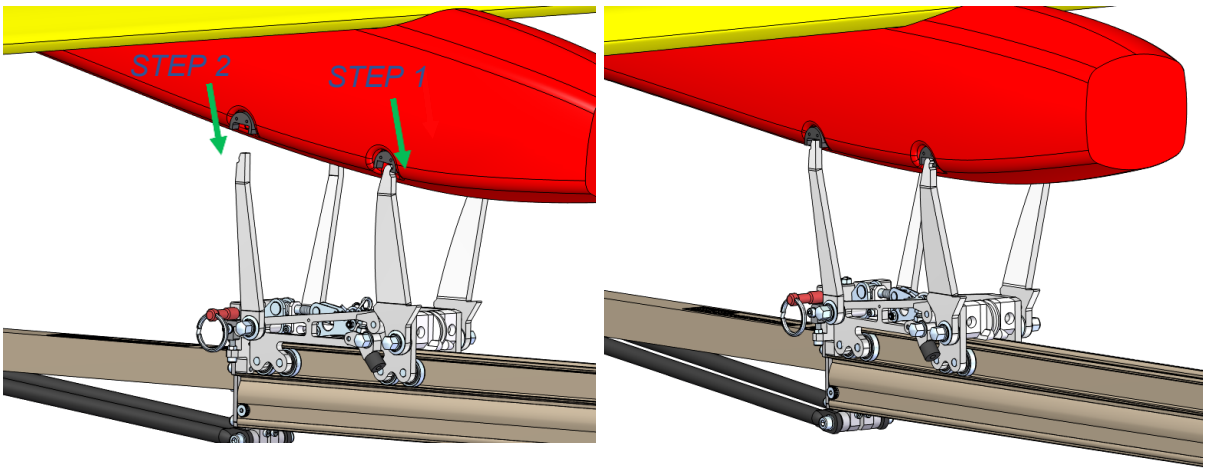


Figure 5-1 *Placing the plane on the launcher*

11. Make sure that carriage safety pin is installed and remote release handle prepared and unengaged. Open slowly the air reservoir outlet valve **(4)** on the compressor to pressurize main valve.

12. When the plane is ready to launch, make sure that all personnel have moved to the safety area **(see Chapter 9, on page 40)**, remove the carriage safety lock pin. For launching move the safety levers up and then press both switches simultaneously. Hold the switches up for at least two seconds. After the start command ends, move the safety levers down again. (**Figure 5-3**).



Figure 5-2 *Connect the plug at the end of the cable to the trigger handle and connect battery*



Figure 5-3 *Trigger handle. For launching move the safety levers up and then press both switches simultaneously. Hold the switches up for at least two seconds. When initiating the start command, a red LED lights up on the remote trigger handle. After the start command ends, move the safety levers down again. The launcher is now assembled and ready for launching.*

NB!
Keep the trigger handle buttons pressed until the plane has left the catapult! Releasing the trigger button closes the main valve and avoids unnecessary pressure loss.

NOTE! If the carriage safety lock pin has not been removed and launch trigger is pressed by chance then you have to start over and repeat all launch preparation steps as described above!

ATTENTION! Keep the safety cover closed at all times to avoid unwanted launcher release. Ignoring that may cause serious injuries.

13. To prepare for the next launch move the carriage to the initial position and secure it with safety lock. Wait for the compressor to build up determined pressure.

    For safety reasons it is recommended to close air reservoir outlet valve between launches to avoid accidental carriage release by pressing the trigger handle buttons.

    It is recommended to change the battery with a fully charged one after the first launch and then change the battery at least after every two consecutive launches.

6 Disassembly procedure



WARNING!
Make sure, that the main valve **(4)** is closed before any further actions. Ignoring that may cause serious injuries.

To disassemble the catapult, perform the following steps:

1. Close the air reservoir outlet valve **(4)**.
2. Turn the compressor OFF.
3. Disconnect pressure delivery tube from the launcher and the main tank.
4. Move carriage to rail rear module (to the area of launch position) and remove the pulling rope from carriage. Remember to put the fastening pin back to its place.
5. Remove ground anchors
6. Open the clamps and dismantle the rail, leaving carriage to the rear rail module.
7. Pack all the parts into their cases**.**

7 Maintenance

7.1 Battery maintenance

Option 1: Makita®'s 18V Li-Ion 6000mAh battery cartridges, Makita® part nr. BL1860B and Makita® battery charger DC18RC or DC18SD.

Charging time for empty battery is about:
a) 55 min for DC18RC charger or b) 130 min for DC18SD charger

NOTE! Battery charger is not provided as voltage and socket may vary by country!

Tips for maintaining maximum battery life:

- Charge the battery cartridge before completely discharged.
- Never recharge a fully charged battery cartridge.
- Overcharging shortens the battery service life.
- Charge the battery cartridge with room temperature at 10˚C - 40˚C (50˚F - 104˚F).
- Let a hot battery cartridge cool down before charging it.

7.2 Launcher maintenance

- Check the pulling rope for broken cords.
- Change the rope if necessary.
- Check that wheels of the carriage are undamaged and are rotating freely.
- Check that the carriage is moving freely on the rail.
- Check that the parts of the rail are clean. Clean if necessary.
- Check that the catapult rail groove (all sides of the groove) where the carriage moves is covered with white lithium grease as needed. We recommend using WD-40 White Lithium Grease.



- Check for the air leakages from tubing between main reservoir and main valve. Tighten the clamps if necessary.
- Check that carriages support levers are undamaged and are moving freely.
- Check that the rail parts are connected tightly. Tighten the rail clamps if necessary.
- Ensure that all drone-side hardpoints are straight, parallel, and securely tightened.



WARNING! Unmaintained launcher may cause launch failures, damage to the equipment and serious injuries.

7.3 Pulling rope replacement

Maintenance instructions

- Pulling rope should be replaced regularly after 100 launches.
- Visual checks should be performed before and after every launch.
- The rope must be replaced immediately when significant wear detected, tear or loose fibers have appeared.
- Make sure that the rope is kept away from mud and dirt during launcher assembling and disassemble.
- The rope replacement can be performed using PL-40 pulling rope replacement kit sold by manufacturer.

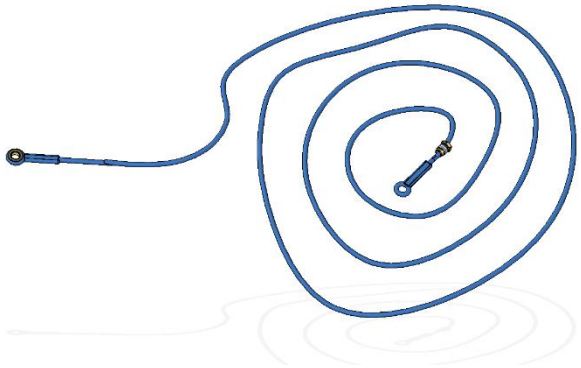


Figure 7-1 *PL-40 pulling rope replacement kit*

WARNING! Unmaintained pulling rope may cause launch failures, damage to the equipment and serious injuries.

Pulling rope replacement instructions

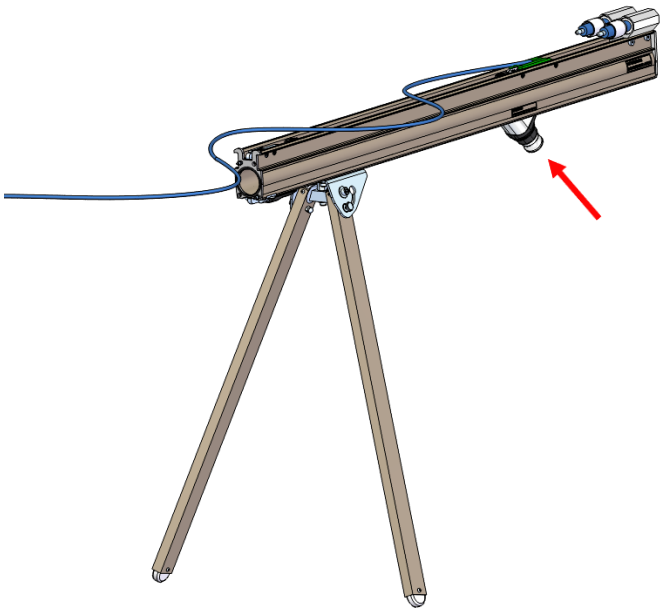


Figure 7-2 *Take the front rail and blow some air into the air intake connector to extract the piston.*

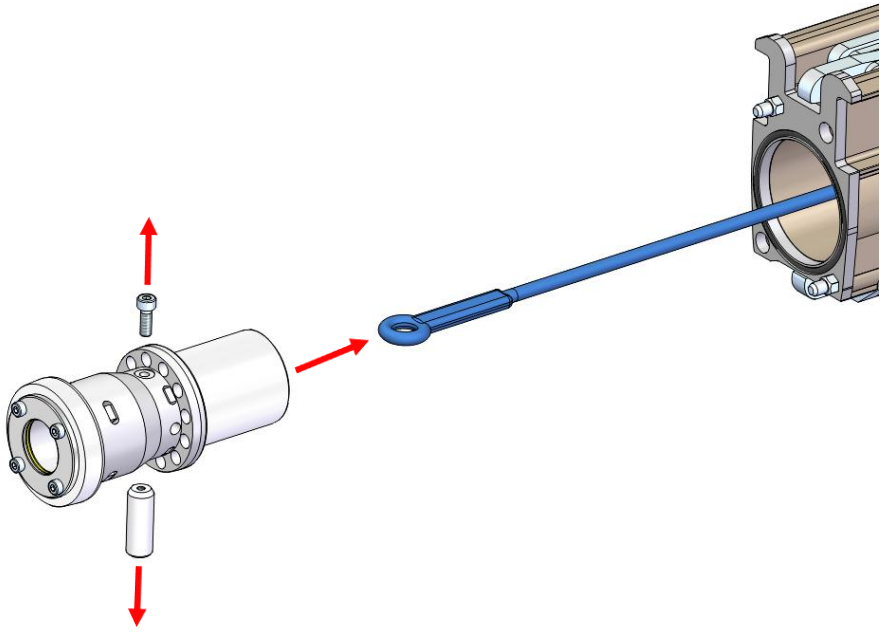


Figure 7-3 *Unscrew the bolt and remove the pin.*

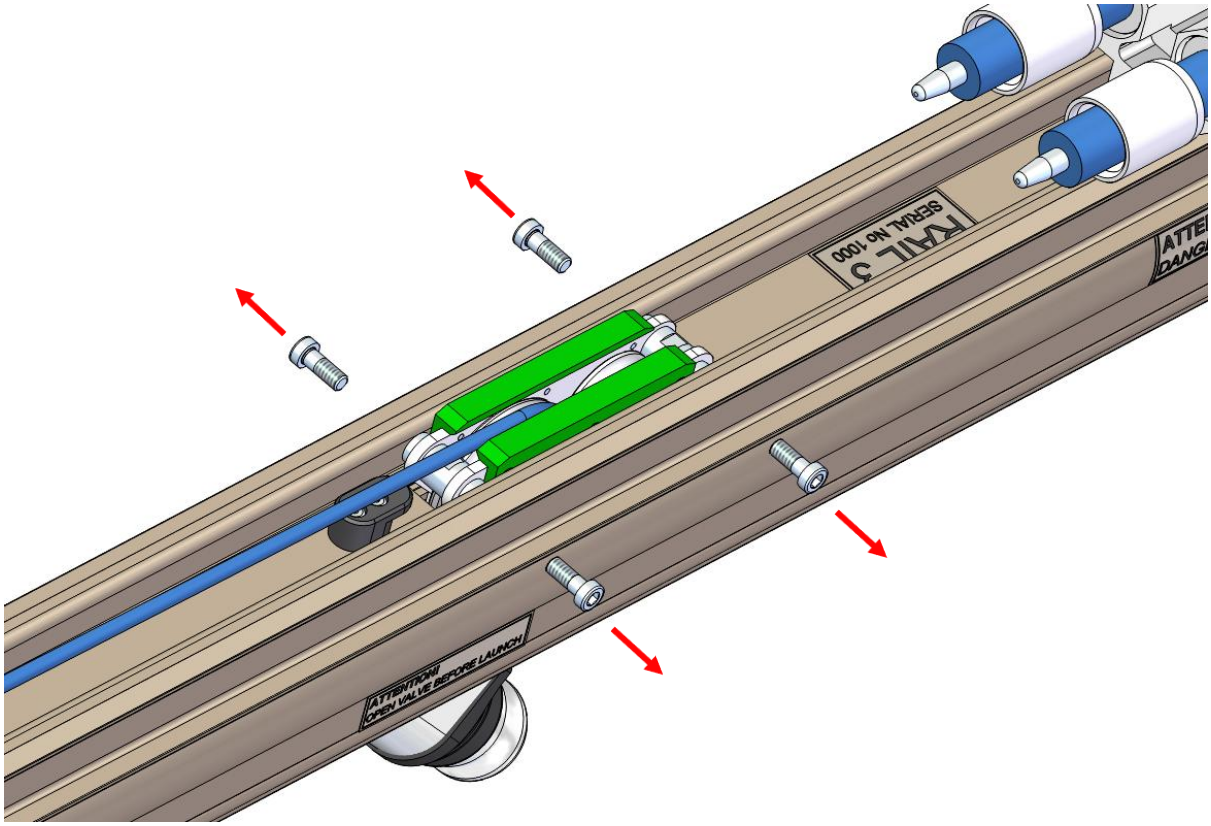


Figure 7-4 Unscrew four bolts holding the roller block.

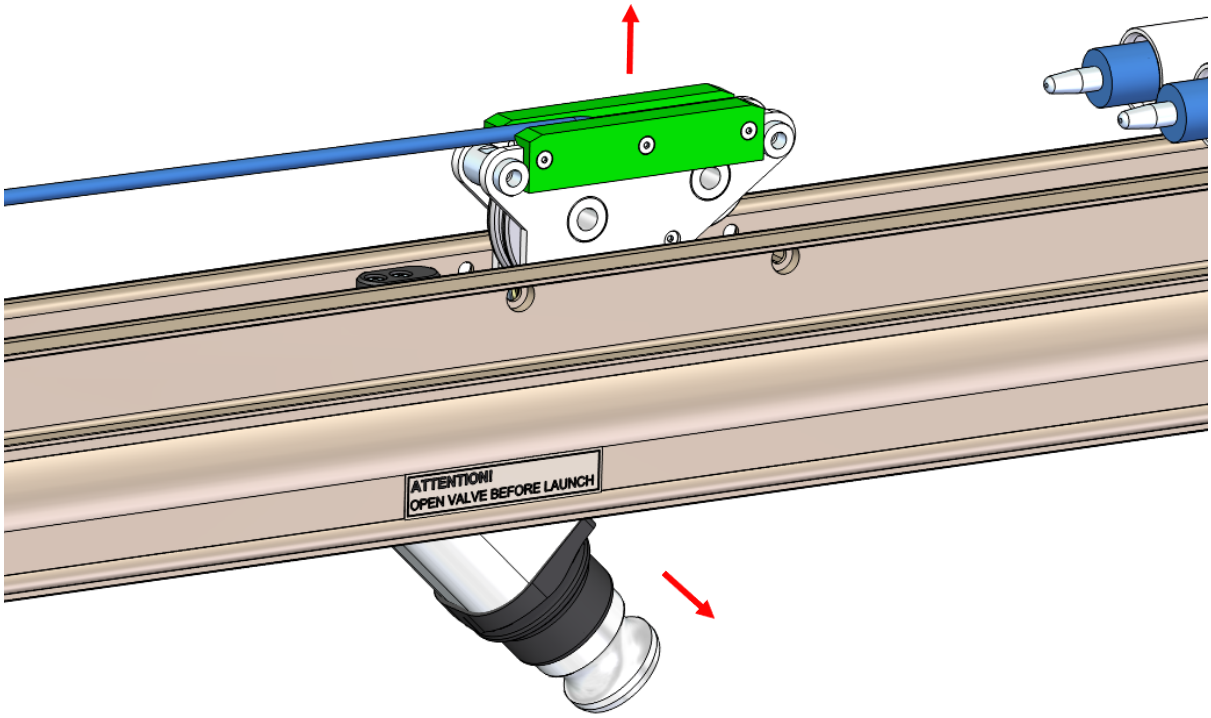


Figure 7-5 Extract the roller block and air supply system.

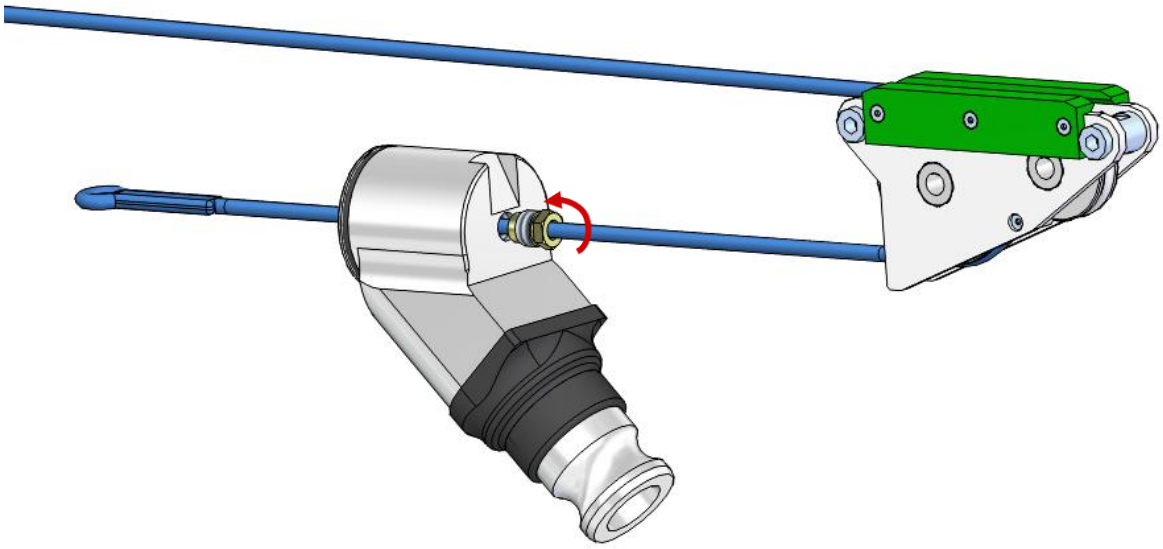


Figure 7-6 *Unscrew the rope guide bushing.*

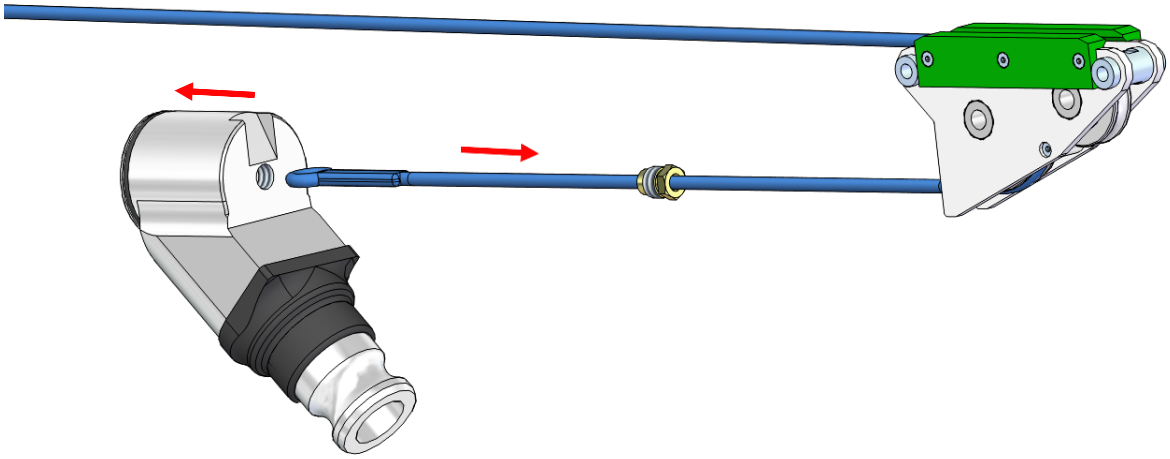


Figure 7-7 Remove the air supply system.

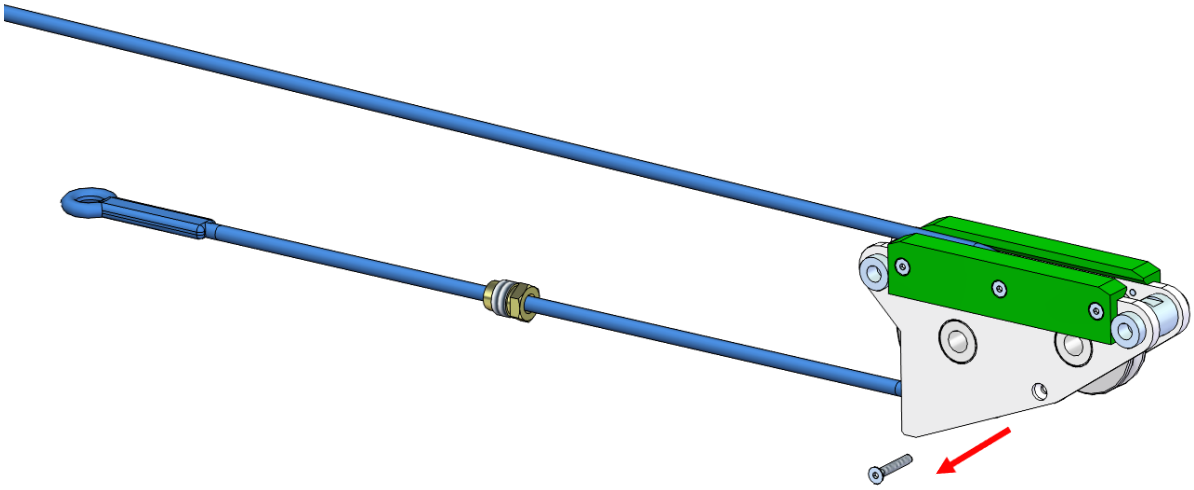


Figure 7-8 *Unscrew bolt holding the side plate.*

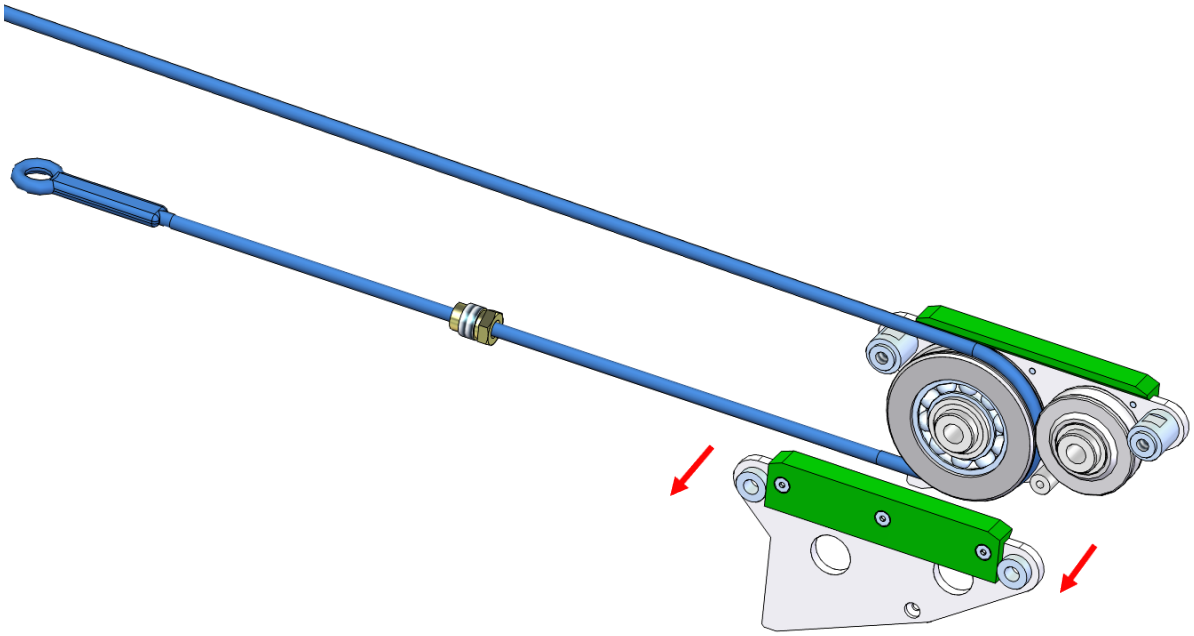


Figure 7-9 *Remove the side plate to gain access to the rope.*

Follow same steps in reverse order to complete the rope replacment.

7.4 Wheel bearing replacement

Maintenance instructions

- Wheel bearings should be replaced regularly after 100 launches.
- Visual checks for signs of wear, damage, or contamination should be performed before and after every launch.
- Keep the bearing area clean and free of contaminants. Dust, dirt, and debris can accelerate wear and damage.
- Use appropriate cleaning methods, such as compressed air or a mild solvent, to remove contaminants from the bearing and track.
- The bearings should be replaced immediately when significant wear, cracks or sealing damage detected.
- The bearing replacement can be performed using carriage replacement kit sold by manufacturer.



1. Wheel bearings – 4 pcs
2. Hex wrench – 1 pc
3. Threadlocker – 1pc

***Figure 7-10** Carriage wheel bearing replacement kit*

Bearing removal

- **Preparation:** Ensure you have the necessary tools and safety equipment, including the hex wrench provided in the kit.
- **Heat the Threaded Area:** Use a heat source to raise the temperature of the area where the wheel bearing shaft is threaded onto the shaft to approximately 250 degrees Celsius. This heating process is essential to soften the thread lock adhesive, making it easier to remove the shaft.
- **Unscrew the Wheel Bearing Shaft:** While heating the threaded area, turn the wheel bearing shaft **clockwise** using the hex wrench. This action will help loosen and remove the shaft from the assembly. The application of heat aids in breaking the adhesive's grip.
- **Remove the Bearing:** Once the shaft is sufficiently loosened, you can remove the wheel bearing from the shaft either by hand or by using a press tool if it is stuck.

ATTENTION! Without applying heat to soften the adhesive before attempting to remove the wheel bearing, there is a higher risk of damaging the bearing shaft's hex slot during the dismantling process.

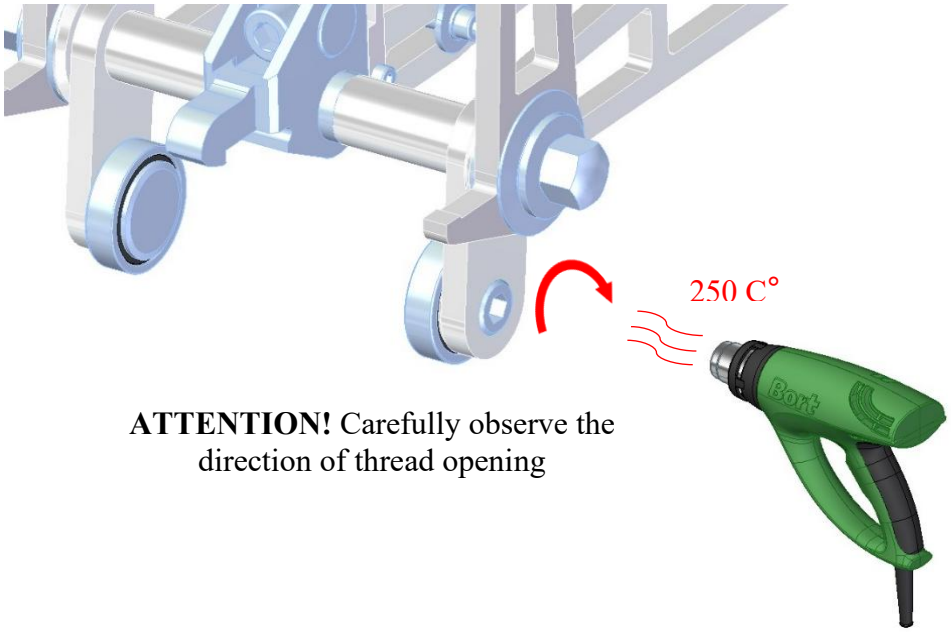


Figure 7-11 *Wheel bearing removal*

Bearing reassembly

- **Preparation:** Preparation: Ensure you have the necessary tools and safety equipment, including the hex wrench and thread lock glue provided in the kit.
- **Assembly:** Install the bearing and support washer as shown in Figure 1-3. Apply the adhesive to the shaft thread and then proceed to reinstall the bearing shaft turning the wrench **counterclockwise**.
- After replacing the bearings, ensure that the sled moves freely on the track

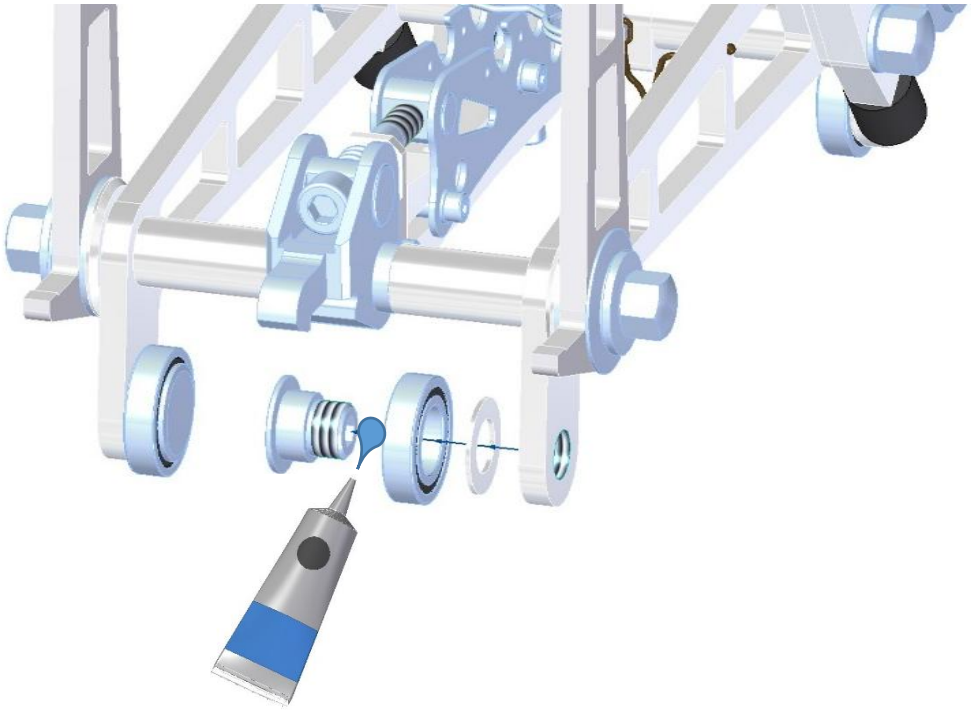


Figure 7-12 *Wheel bearing reassembly*

8 Technical Specifications

- Launch angle — 12 degrees
- Overall rail length when assembled — 5m
- Rail length when disassembled — 1,28 m
- Weight of catapult set, 2 boxes — Total 68 kg
- Launcher case external dimensions — 1480x440x170mm 43kg
- Compressor case external dimensions — 360x360x680mm 25kg
- Maximum plane mass — 30 kg
- Maximum allowed launch velocity — 25 m/s (meter per second)
- Puller configuration aircraft — YES
- Pusher configuration aircraft — YES
- Set-up time — 5min
- Working temperature — -20…. +50C°



ATTENTION! It is user responsibility to maintain relationship between plane mass and determined launch pressure so, that maximum launch velocity 25 m/s is not exceeded.

Table 2 Approximate launch velocities versus plane mass and launching pressure.

Pressure (bar)	Plane mass (kg)			
	10	15	20	30
5	18,4	15,1	13,1	10,7
6	20,2	16,5	14,3	11,7
7	21,8	17,9	15,5	12,6
8	23,3	19,1	16,5	13,5
9	24,8	20,3	17,6	14,3
10	26,1	21,3	18,5	15,1
11	-	22,4	19,4	15,8
12	-	23,4	20,3	16,5
13	-	24,3	21,1	17,2
14	-	25,3	21,9	17,9
15*	-	26,1	22,7	18,5
16*	-	-	23,4	19,1
Actual launch speeds may vary depending on the wind speed, aircraft engine propulsion force and proper support by grounding anchor.				
* Auxiliary pressure source up to 16bar needed. Contact seller for more details.				

COMPRESSOR UNIT
Quick guide

ATTENTION! Do not handle any pneumatic tubing until making sure that the valve of the main pressure tank is closed!

ATTENTION! Carriage has to be secured by the safety lock pin before opening the main valve!

WARNING! Open the main valve only before launch and close it after launch as soon as possible! Ignoring that may cause accidental sled or pressure release and lead to serious injuries or death!

Normal operation:
For switching the compressor unit ON or OFF, press **ON button**.

Setting the target pressure (SET MODE):
For adjusting the target pressure value make sure, that the system is switched OFF. Hold **SET button** and then start the system by pressing **ON button**. Release both buttons and wait until the target pressure is reached. Press **SET button** to save the target pressure value into memory or **ON button** to switch the system OFF and maintain previous value.

Status LED states (illuminated push-buttons)

PWR LED	SET LED	MOTOR STATE	SYSTEM STATUS
○	○	**OFF**	Turned OFF
●	●	**OFF**	Target pressure reached
●	✹	**OFF**	Pressure in the tank over the target pressure
●	●	**ON**	Pressurizing to target pressure
●	○	**ON**	SET MODE (Pressurizing tank)
●	○	**OFF**	SET MODE (Maximum pressure reached to 14 bar)
✹	**X**	**OFF**	Battery voltage too low

● - LED on ○ - LED off ✹ - LED blinking **X** - Any state

ATTENTION! Before launching the plane always check the pressure on the manometer, making sure, that the target pressure is achieved.

!!!READ USER MANUAL FOR DETAILED INFORMATION!!!

LAUNCH PROCEDURE
Quick checklist

For ensuring safe launch procedure perform the following steps:

- ☐ Unpack and assemble the rail.
- ☐ Make sure, that the carriage is in its initial position and secured with safety lock pin.
- ☐ Make sure that the pulling rope is attached to the carriage and installed properly.
- ☐ Make sure, that the compressor unit ON/OFF switch is in OFF position and the main reservoir outlet valve is in closed position.
- ☐ Install pressure hose connecting main valve outlet and the launcher inlet. Make sure that the clamps are perfectly locked and securing ring also locked.
- ☐ Insert a fully charged battery cartridge into its connection slot on the compressor unit.
- ☐ Turn on the compressor unit by pressing ON button and wait until target pressure is reached.
- ☐ Check the pressure manometer on the pressure regulation valve to assure that the determined pressure is achieved.
- ☐ Make sure that carriage safety pin is installed and remote release handle prepared and unengaged.
- ☐ Place the plane on the launcher. Make sure that all four support levers are correctly inserted into throwing slots.
- ☐ When the plane is ready to launch, make sure that all personnel have moved to the safety area
- ☐ Open slowly the main outlet valve on the compressor unit to pressurize main valve.
- ☐ Remove the carriage safety lock pin. To launch, open the safety covers on the trigger handle buttons and press both buttons simultaneously to open the main valve.
- ☐ For safety reasons it is recommended to close air reservoir outlet valve between launches to avoid accidental carriage release by pressing the trigger.



ALWAYS USE GLOVES AND GOGGLES when servicing the plane or the launcher!



NEVER LAUNCH WITHOUT PAYLOAD as it will damage the device and surrounding persons!



NEVER LAUNCH WITHOUT SECURING LAUNCHERS REAR LEG TO THE GROUND



ATTENTION! Before launching make sure that all team members are in safety area behind the launcher!



ATTENTION! Carriage has to be secured by the safety lock pin before opening the main valve!



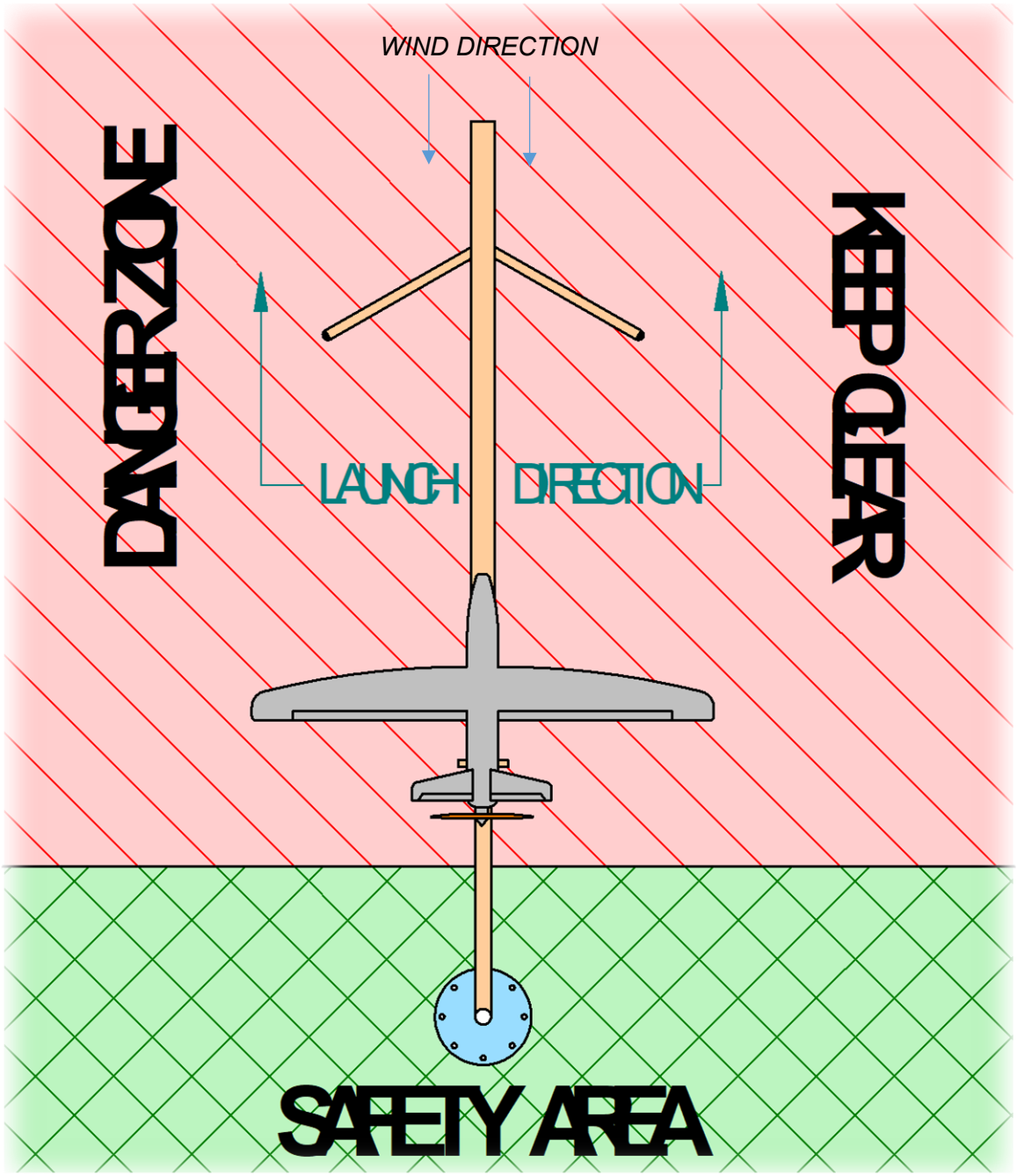
WARNING! Open the main valve only before launch and close it after launch as soon as possible! Ignoring that may cause accidental sled or pressure release and lead to serious injuries or death!



ATTENTION! Remove the carriage safety lock only just before the launch!

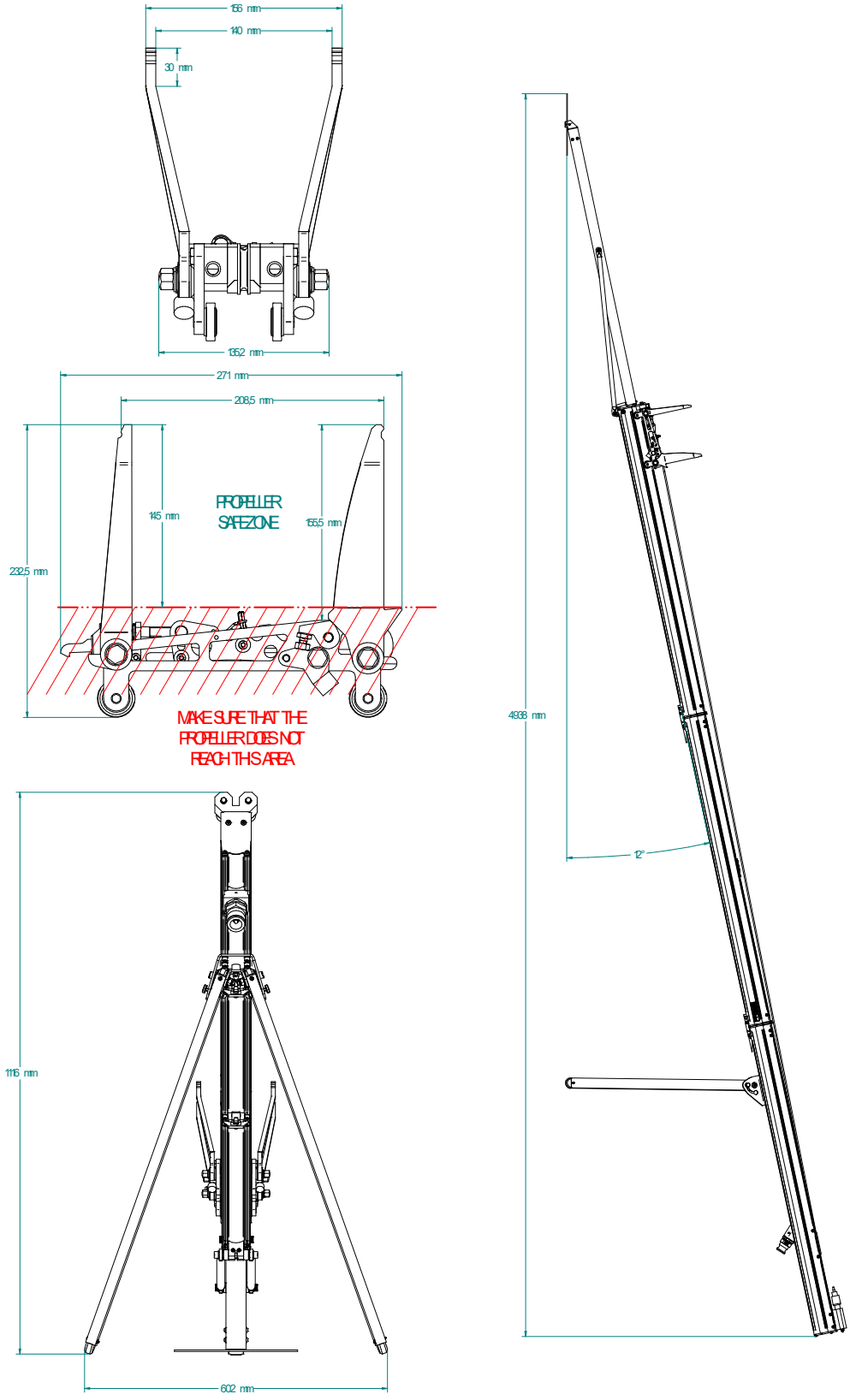
!!!READ USER MANUAL FOR DETAILED INFORMATION!!!

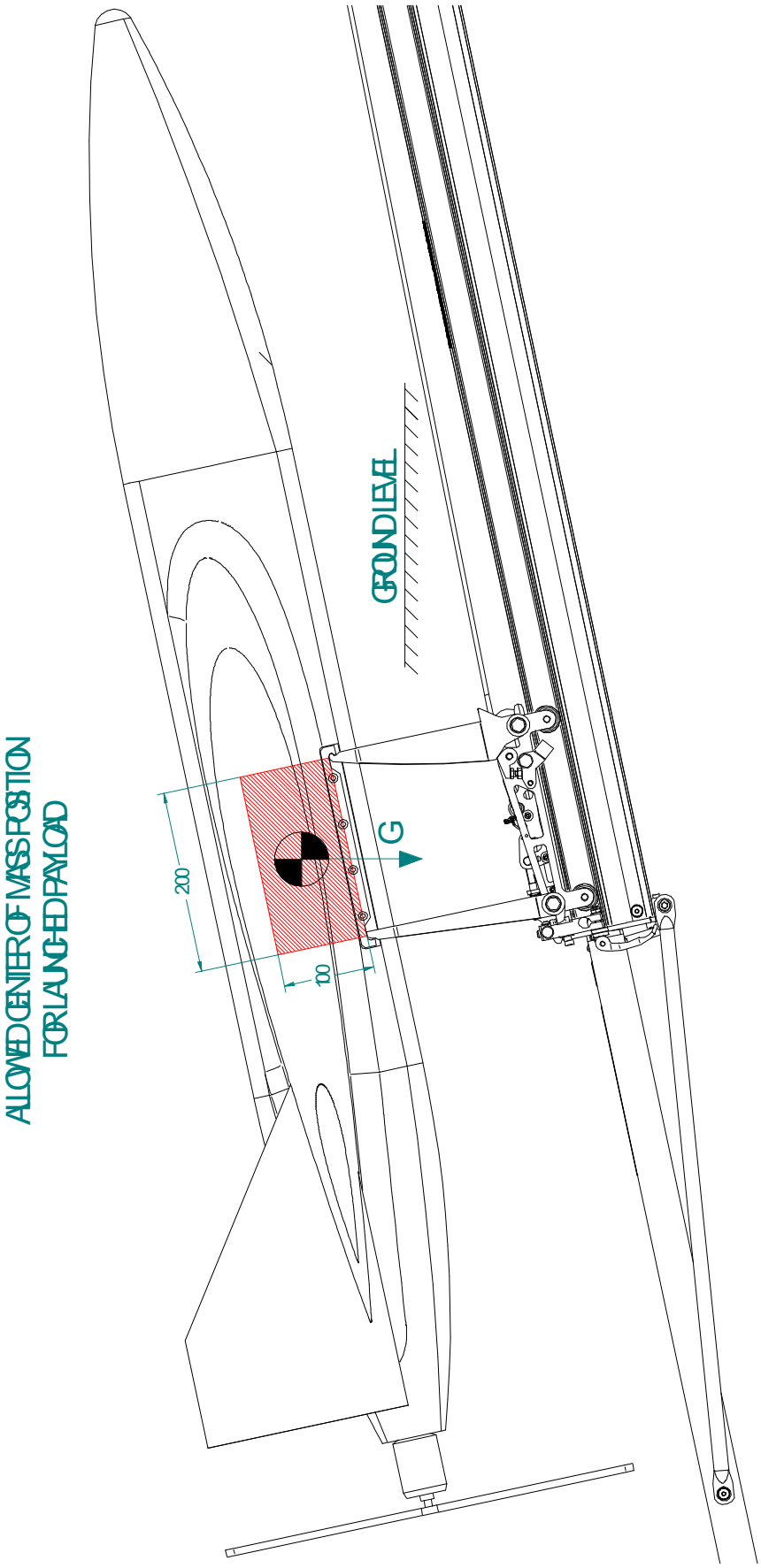
9 Safety area schematics



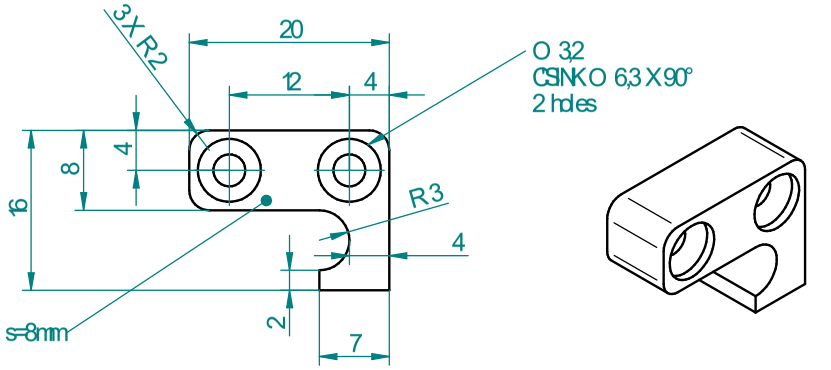
ATTENTION! Before launching make sure that all team members are in safety area behind the launcher!

10 Plane installation documentation in standard configuration



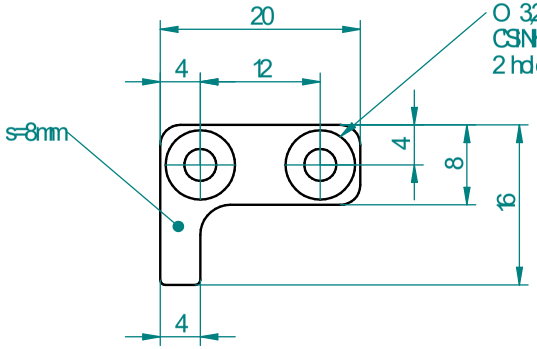

ALLOWED CENTER OF MASS POSITION
FOR LAUNCHED PAYLOAD
GROUND LEVEL
G
200
100

Rev.No	Revision note	Date	Signature	Checked

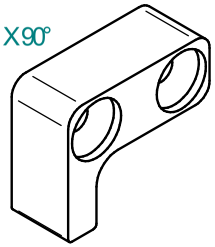


O 3,2
CSINK O 6,3 X 90°
2 holes

FRONT RIGHT CONNECTOR



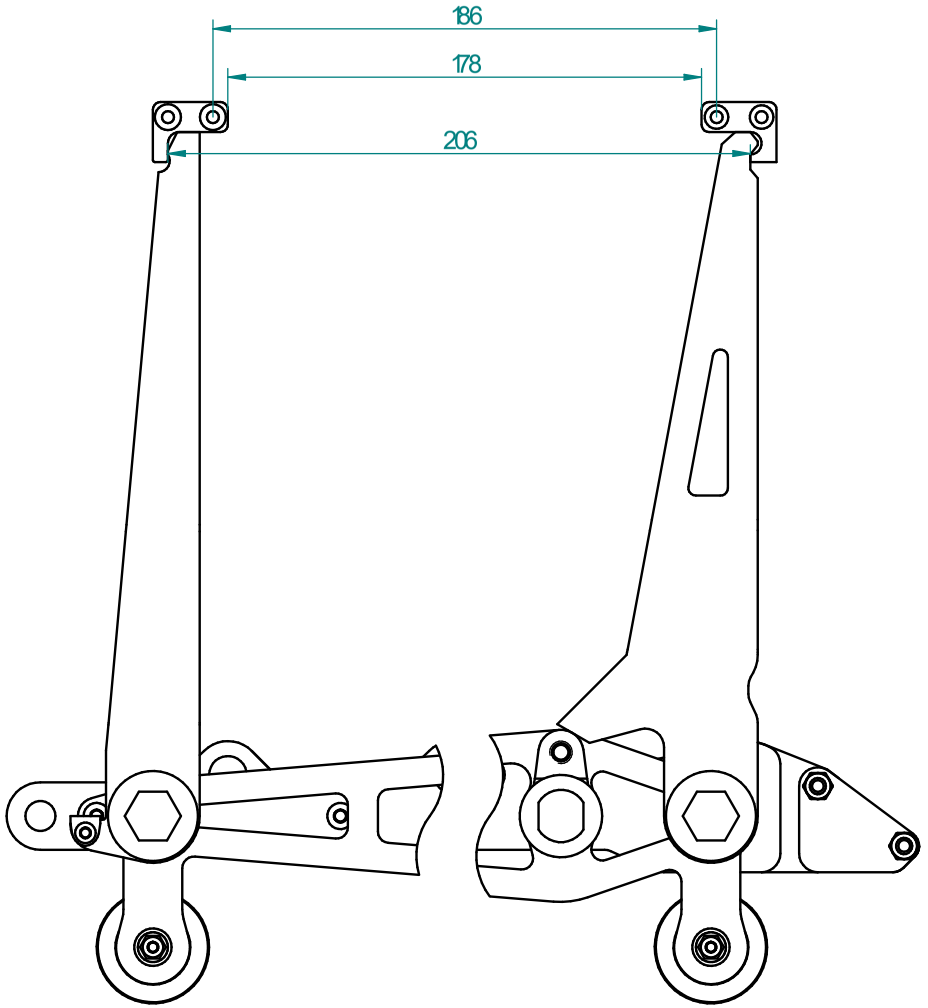
O 3,2
CSINK O 6,94 X 90°
2 holes



REAR RIGHT CONNECTOR

NOTE: LEFT SIDE PARTS ARE MIRRORS OF THE RIGHT SIDE PARTS

LAUNCH DIRECTION



1. Material: Aluminum, 6061-T6
2. Surface area: 1117,53 mm^2
3. Coating: Natural anode

This drawing the owner of which we are, is furnished upon express condtion that it must not be copied or in any other way directly or indirectly used detrimental to our interests. ELI Llc.

Seda joonist ei tohi ilma meie rõusolekuta kopeerida ja muul viisil otseselt ega kaudselt kasutada, kui see võib kahjustada meie huvisid. ELI OÜ

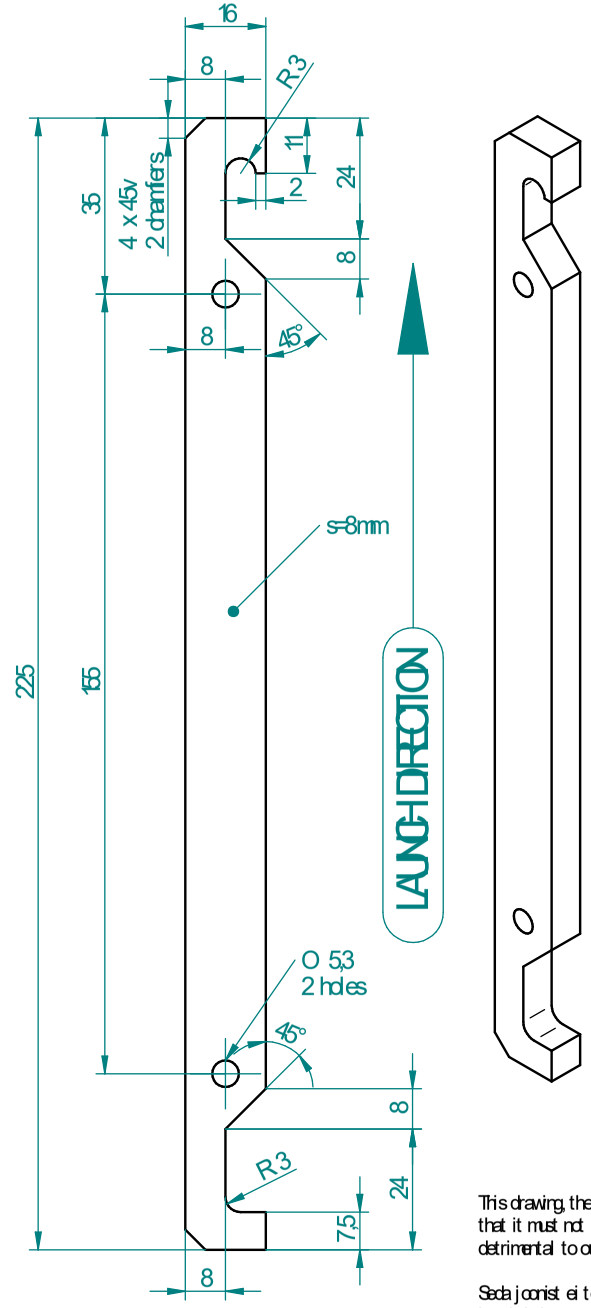
Designed by	Checked by	File name	Date	Mass	Scale
ELI Ltd.	Jaan Palm		4.04.2016	0,004 kg	

Dimensions are in millimeters
Unshown tolerances: ISO 2768-m
Unshown raduses: R=0,2mm
Unshown surface roughness: Ra 6,3

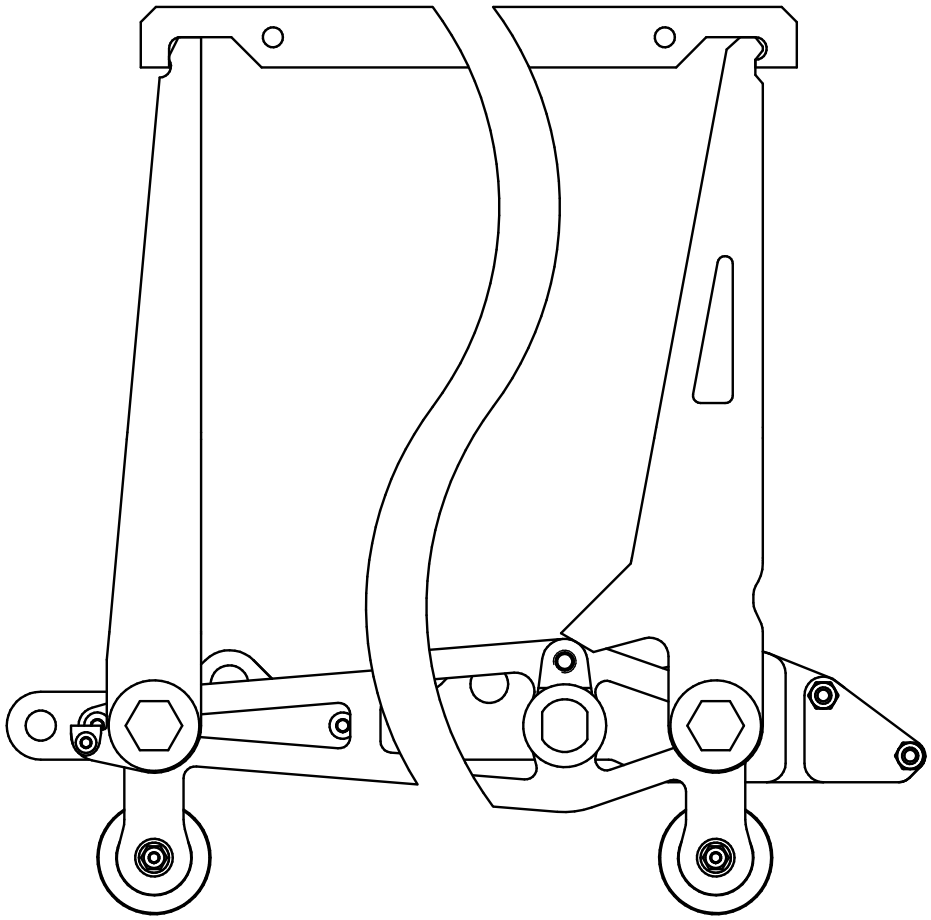
Plane connection hook option A

Edition	Sheet
A	1 of 1

Rev.No	Revision note	Date	Signature	Checked



LAUNCH DIRECTION



This drawing, the owner of which we are, is furnished upon express condition that it must not be copied or in any other way directly or indirectly used detrimental to our interests. ELI Llc.

Seda joonist ei tohi ilma meie nõusolekuta kopeerida ja muul viisil otseselt ega kaudselt kasutada, kui see võib kahjustada meie huvisid. ELi OÜ

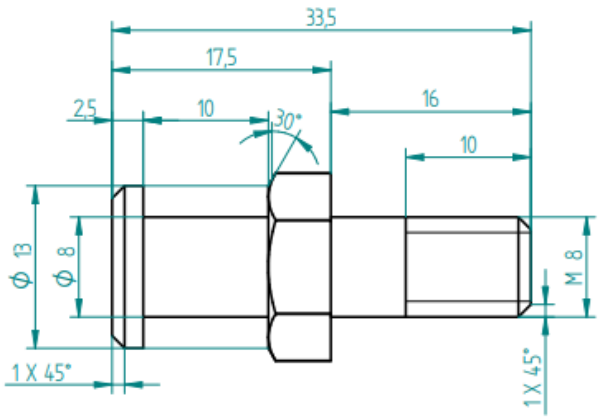
Designed by	Checked by	File name	Date	Mass	Scale
ELI Ltd.	Jaan Palm		4.04.2016		

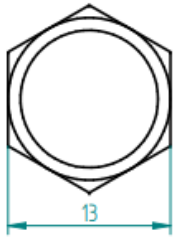
Dimensions are in millimeters
Unshown tolerances: ISO2768-m
Unshown raduses: R-0.2mm
Unshown surface roughness: Ra 6.3

Plane connection hook option B

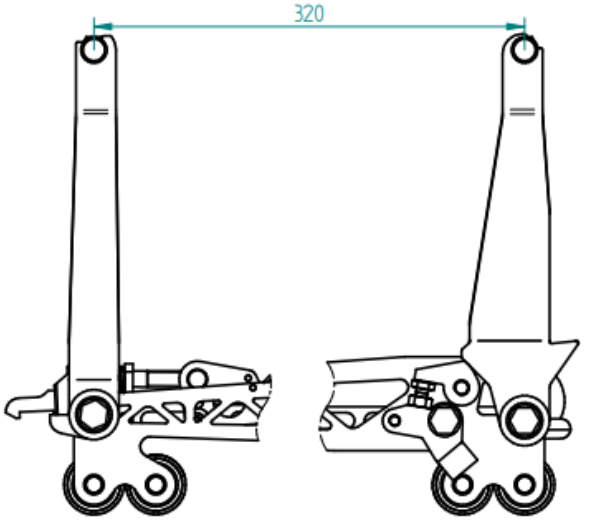
Edition	Sheet
A	1 of 1

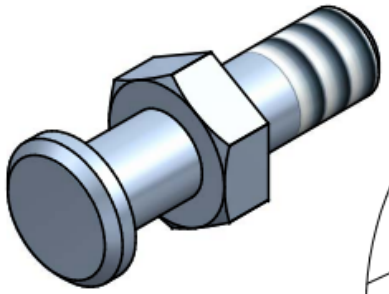
RevNo	Revision note	Date	Signature	Checked

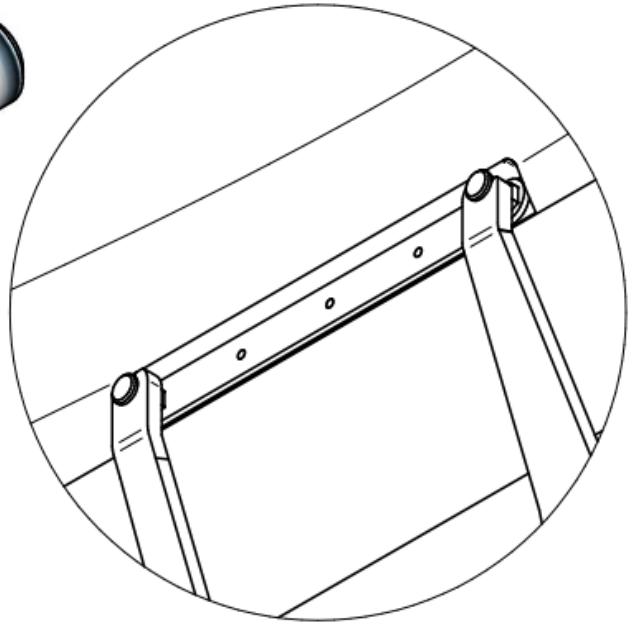


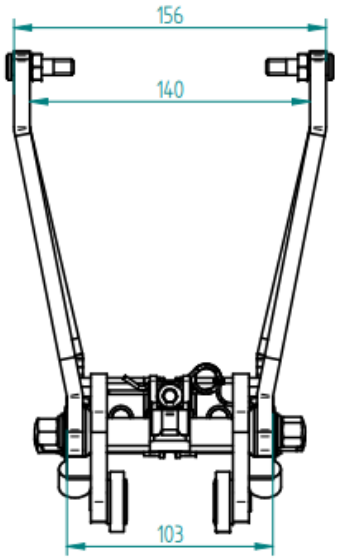


LAUNCH DIRECTION









1. Material: AISI 304
2. Surface area: 1390,16 mm^2
3. Coating: Natural anode



This drawing, the owner of which we are, is furnished upon express condition that it must not be copied or in any other way directly or indirectly used detrimental to our interests. ELI LLc.

Seda joonist ei tohi ilma meie nõusolekuta kopeerida ja muul viisil otseselt ega kaudselt kasutada, kui see võib kahjustada meie huvisid. ELI OÜ

Designed by	Checked by	File name	Date	Mass	Scale
ELI Ltd.	Jaan Palm		06/10/2025		

Dimensions are in millimeters
Unshown tolerances: ISO 2768-m
Unshown radiuses: R=0.2mm
Unshown surface roughness: Ra 6.3

Plane connection hook option C

Edition	Sheet
A	1 of 1